Group flights in free flight

Delayed contacts, groups and episodes:
a method to label every instant of every flight

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Abstract

To compare how paraglider pilots move in a group and alone, we must know, at every instant of every flight, whether the pilot flies in a group and with whom. Three difficulties stand in the way, and the method answers each with one idea. Pilots who fly together follow the same route minutes apart: two pilots are in contact when one of them is where the other was, or will be, a few minutes earlier or later, within a flat box of tolerances and with headings compared only in glide, after the following with a local delay of Li, Wu and Crofoot [16]. Contact is not transitive and changes in time: at each instant the configurations are the connected components of the contact graph, organised in groups and episodes as in the grouping structure of Buchin et al. [7]. Pilots also meet by chance on the same ridges and in the same thermals: copies of the flights shifted in time measure chance on the data, and a contact counts only if it lasts long enough for at most one in ten to be chance. On the FFVL archive the delay is measured at 2 min and a contact must last 38 min; 21.7% of the paraglider flights and 6.9% of the hang-glider flights then fly in a group at some point, in 17 169 episodes. The groups are mostly pairs; their pilots keep the same pace with no extra pull between them, their sizes change one pilot at a time, and pilots return to the same partners more than chance explains. The output is a few tables that label every instant of every flight.

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1 Introduction

The thesis studies how pilots move in cross-country flight: how long they climb in thermals, how far they glide, how their mean squared displacement grows. These statistics are usually estimated as if each flight were an independent sample. The assumption fails when pilots fly together: they share thermals and routes, so their tracks are correlated, and the group may change the behaviour itself. In soaring birds, social information on the position of thermals changes the gliding strategy [31], and in a migrating flock a few individuals find the thermals while the others follow them [9, 20]. To compare group flight with solo flight we must know, at every instant of every flight, whether the pilot flies in a group and with whom. This note gives a method that answers, and applies it to the FFVL archive.

1.1 What flying together means

The method makes operational an intuitive notion of flying together (Figure 1):

(R1) the pilots take off from the same area, possibly from different takeoffs;

(R2) their departures may be staggered, as long as they keep each other in sight;

(R3) they follow roughly the same route: nearby places and similar motion; the tracks need not pass through the same points;

(R4) the composition of the group changes in time: some pilots join, others change route.

1.2 Three difficulties, three ideas

Three simple rules miss these requirements, each on one difficulty (Figure 2). Each difficulty gives one idea of the method.

1. Together, minutes apart. Pilots on one route pass the same places minutes apart, and two minutes are about a kilometre. Rules of strict synchrony (flock [11], convoy [12], swarm [15]) ask for proximity at the same instant and never see them together (Figure 2b); the swarm admits non-consecutive instants, but no delay between members. *Idea:* two pilots are in contact when one of them is where the other was, or will be, at most a few minutes earlier or later. This is the following with a local delay of Li, Wu and Crofoot [16], which we adapt to soaring in three ways: the tolerance is a flat box, wide horizontally and narrow in height, like a group;

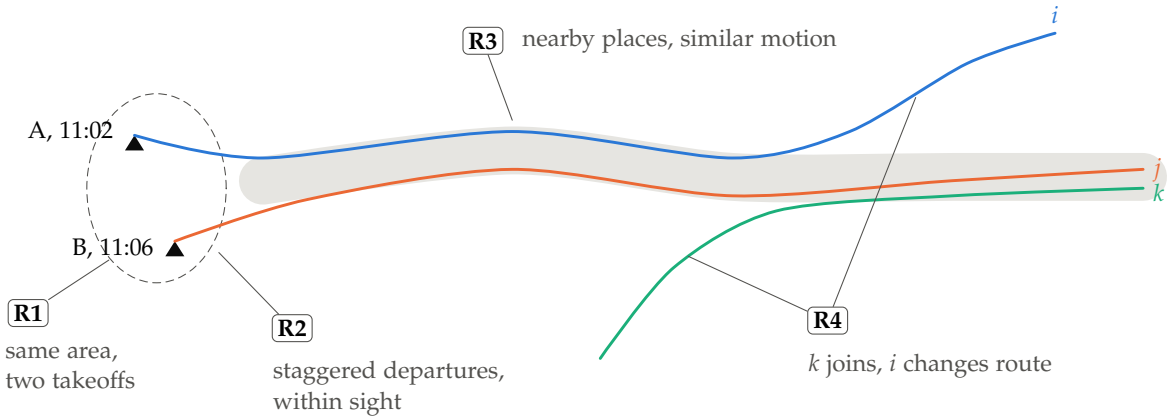


Figure 1: The four requirements on a schematic map. Pilots i and j take off from two places of the same area four minutes apart and follow the same corridor (grey) without flying over the same points; k joins later and i leaves the corridor.

headings are compared only in glide, where they show the route; the takeoff area, where everyone is close to everyone, is left out (Section 3).

2. Not transitive, and changing in time. If 1 flies with 8 early in the afternoon and 8 with 14 later, 1 and 14 may never have met; and pilots join and leave all the time (R4). A window on the departures, such as "departures within T minutes", makes groups that depend on the reference pilot (Figure 2c). *Idea:* groups are defined instant by instant, as the connected components of the graph of contacts: each component, with its edges, is a *configuration*. A fixed set of pilots that stays one component is a *group*, and a chain of groups that share members is an *episode*, after the grouping structure of Buchin et al. [7]. Episodes tie the flights into *dependency blocks*, which the statistics resample together (Section 4).

3. Chance meetings. Routes follow the terrain and many thermals are anchored to it: independent pilots from the same site fly the same ridges and climb in the same thermals. Geometric similarity of the tracks (Hausdorff, Fréchet, unconstrained DTW; survey in [25]) finds such tracks similar even hours apart (Figure 2a); unconstrained DTW also warps time without limit and loses the delay that (R2) bounds. *Idea:* chance is measured on the data. Copies of the flights shifted in time keep the routes and lose the synchrony, so their contacts are chance. A contact counts only if it lasts at least τ^*, chosen so that at most one contact in ten is chance: $\tau^* = 38.1$ min on the archive (Section 5).

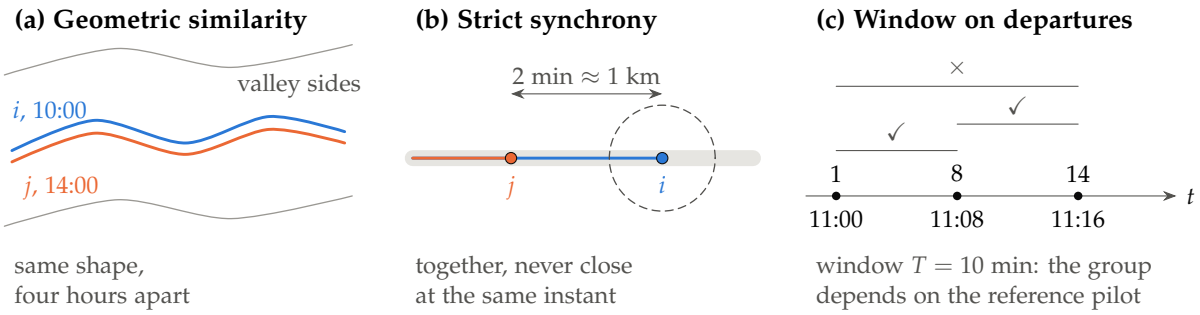


Figure 2: Why three immediate rules fail. (a) Two independent pilots on the same valley produce similar tracks hours apart. (b) Two pilots on the same route two minutes apart never enter each other's proximity disc. (c) With a window of 10 minutes, 1–8 and 8–14 pass and 1–14 fails.

1.3 The method in six steps

The three ideas chain into six steps, from two tracks to the units of the statistics (Figure 3); Table 1 names each object in plain words, with the section that defines it. Figures 5 and 15, marked ▶ , are animated: in Adobe Acrobat Reader, or another viewer that runs the JavaScript of PDF files, the buttons below them play the animation; other viewers show the figure as printed. The same animations are GIF files in the folder `animations/`.

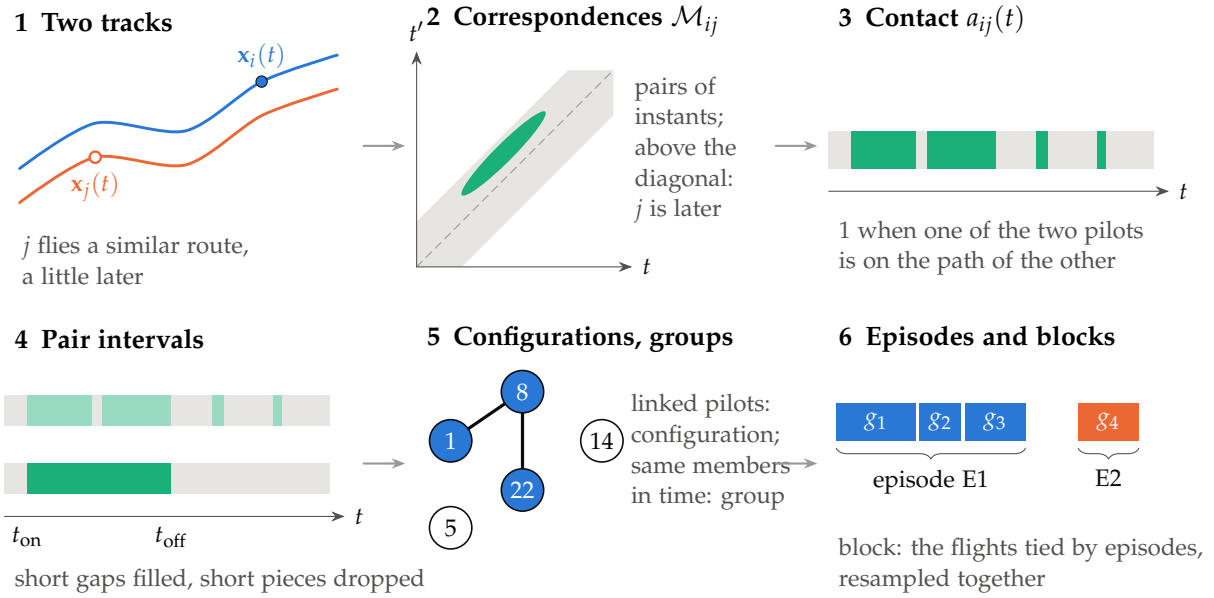


Figure 3: The method in six steps. Panels 1–3: the pair contact (Section 3). Panels 4–6: from pairs to groups (Section 4). Flights 1, 8, 22, 14 and 5 are the example day of Figure 15.

Object	In words	Symbol	Section
Departure	the first fix farther than R_{dep} from the takeoff	k_i^{dep}	3.1
Correspondence	two instants of the same clock, t for i and t' for j, at most δ apart, at which the two pilots pass the same place within the tolerances	$(t, t') \in \mathcal{M}_{ij}$	3.2
Contact	at instant t, one of the two pilots is on the path of the other	$a_{ij}(t) = 1$	3.2
Pair interval	a stretch of contact, with short gaps filled and short pieces dropped	$[t_{\text{on}}, t_{\text{off}})$	4.1
Configuration	the pilots joined to each other by chains of contacts at instant t, with their edges; a new edge gives a new configuration	component of $G(t)$	4.3
Group	a fixed set of flights that stays one component over a maximal interval	g	4.3
Membership	the label of flight i at instant t: local phase, solo, or its group	$\gamma_i(t)$	4.3
Episode	a chain of groups, each starting when the previous one ends and sharing a member with it		4.4
Dependency block	the flights tied together by episodes; the bootstrap resamples them together		4.5

Table 1: A minimal dictionary of the objects of the method.

1.4 What we take and what is new

Each ingredient of the method has a precedent (Appendix A, Table 9). The pair contact is the following with a local delay of Li et al. [16]; the group structure is that of Buchin et al. [7]; groups built from delayed relations exist in mFLICA [4], directed, with leaders and on sliding windows; the time shift of the null model comes from Spiegel et al. [24]. What we did not find in the literature is:

(i) the Reeb structure of [7] built on contacts with a local delay instead of simultaneous proximity;

(ii) the adaptations to thermal flight: a vertical tolerance that separates pilots circling in the same thermal at distant heights, and headings compared only in glide (Section 3.3);

(iii) the application to a free-flight dataset of this size, with thresholds calibrated on a null model.

The contribution is therefore a physically motivated adaptation and a scalable implementation, more than a new method in itself, and the thesis should present it in these terms.

1.5 Plan of the note

Section 2 fixes the data and the notation. Section 3 defines the contact of two pilots, Section 4 builds the groups from it, and Section 5 defines chance and the minimum duration τ^*. Section 6 sets the parameters: measured on the data where the data fix them, declared as definitions where they do not. Section 7 reports what we find on the archive: what the groups are, what holds them together, who leads them and with whom pilots fly. Section 8 says how the labels serve the comparison of group and solo flight, and lists the choices to declare and the limits. The appendices hold the literature (Appendix A), the delay of a contact that says who follows whom (Appendix B), the algorithm, with the proof that its filter loses no contact (Appendix C), the output tables (Appendix D), why the two halves of the contact differ (Appendix E), the terms of graph theory (Appendix F), two models (Appendix G) and further evidence (Appendix H).

2 Data and notation

What does the method read, and in which frame?

Flights. The flights are those of the xc-dynamics pre-processing. Each flight $i \in V$ is one or more segments of fixes, from its takeoff k_i^0 to its landing k_i^{fin} (step 8 of the pre-processing); between two segments the track has a gap. A fix has a time, a latitude φ, a longitude λ and the GNSS altitude z, at the interval of its logger: one second for 70% of the paraglider flights, two to ten seconds for most of the others. Time k is in UTC seconds from the start of the day, one clock for all flights; IGC times are whole seconds.

Frame. As in the thesis (chapter Trajectory pre-processing, section Local coordinates and vertical datum), each fix is moved to the point of the WGS84 ellipsoid below it, $h = 0$, and converted to Earth-centred Cartesian coordinates, $\mathbf{x}_i(k) \in \mathbb{R}^3$. Here they are not rotated into the east and north of each flight, so all flights share one frame. The distance between two positions is the chord of the ellipsoid, shorter than the distance along it by about $d^3/24R^2$: a millimetre at $d = 10$ km. The altitude is z as recorded.

Phases. The phases of this run are Vilpellet's, computed by the old repository (soaring, tag `legacy-final`) on the flights logged every second only, 70% of the paraglider flights; the phase labelling of xc-lab (note *Flight phases with a hidden Markov model*), meant for loggers of 5 s or less, was not used. A labelled fix has one of three states, $s_i(k) \in \{\mathsf{climb}, \mathsf{search}, \mathsf{transition}\}$: climbing in a thermal, searching for the next one, gliding between two thermals; *glide* means transition. The fixes of the other flights and of pieces too short to be labelled stay unclassified.

Table 2 collects the symbols and the parameters.

What we keep. The positions of all flights in one Earth-centred frame, the recorded altitude, and the phase of flight on the flights logged every second.

Symbol	Meaning	Start	Chosen
$\mathbf{x}_i(k) \in \mathbb{R}^3$	position: the point of the ellipsoid below the fix, Earth-centred		
$z_i(k)$	GNSS altitude		
$s_i(k)$	state: climb, search, transition (glide), or unclassified		
$\hat{\mathbf{v}}_i(k)$	horizontal heading, used only in glide		
k_i^0, k_i^{fin}, k_i^{dep}	takeoff, landing, departure (Section 3.1)		
T_i^{dep}	the grid instants from departure to landing at which flight i has a position		
R_{dep}	radius of the local phase around the takeoff	3 km	3 km
Δ	step of the common time grid $\Delta\mathbb{Z}$	5 s	5 s
$h_{\max}$	longest time between two fixes across which a grid position is interpolated	10 s	10 s
δ	maximum delay between two pilots, always a time	5 min	2 min
ε_h, ε_z	horizontal and vertical tolerances	500 m, 300 m	500 m, 300 m
$\theta_{\max}$	maximum angle between the headings in glide	45°	45°
g	longest gap filled in a contact	120 s	120 s
τ^*	minimum duration of a contact	from the null model	38.1 min
v_{xc}	mean cross-country speed, measured on the data	≈ 5 m/s	
$v_{\max}$	top speed between two fixes, set by the pre-processing (`speed.max_kmh`)	100 km/h	100 km/h
$\varepsilon_{\mathrm{vis}}$	visual range	1.5 km	
$r_{\min}$, R_s	smallest time shift and radius of the takeoffs in the null model	$> 2\delta$, 10 km	$2\delta + \Delta$, 10 km
q	false discovery rate accepted for the contacts	0.05	0.1

Table 2: Symbols and parameters, with their starting values and the values chosen on the data (Section 6). Beware: in [30] δ is a tolerance on indices, in [13] a distance threshold, in [7] a minimum duration; here it is always a delay.

3 The pair contact

When are two pilots together? When one of them is where the other was, or will be, at most δ earlier or later (Section 1.2). This section defines that contact and argues each of its terms.

3.1 Local phase

Near the takeoff every pilot is close to every other, ridge soaring or waiting for the thermals, and a contact there says nothing about the route (R3). We start looking at the departure, the first fix farther than R_{dep} from the takeoff:

$$k_i^{\text{dep}} = \min\{k : \ \|\mathbf{x}_i(k) - \mathbf{x}_i(k_i^0)\| > R_{\text{dep}}\}. \tag{1}$$

From the departure to the landing the flight is evaluated on T_i^{dep}, the instants of a common grid, one every Δ seconds, at which it has a position (Appendix C.2). The instants before the departure, and the whole of a *local* flight that never leaves the disc, get the state L and do not count as solo flight (Figure 4).

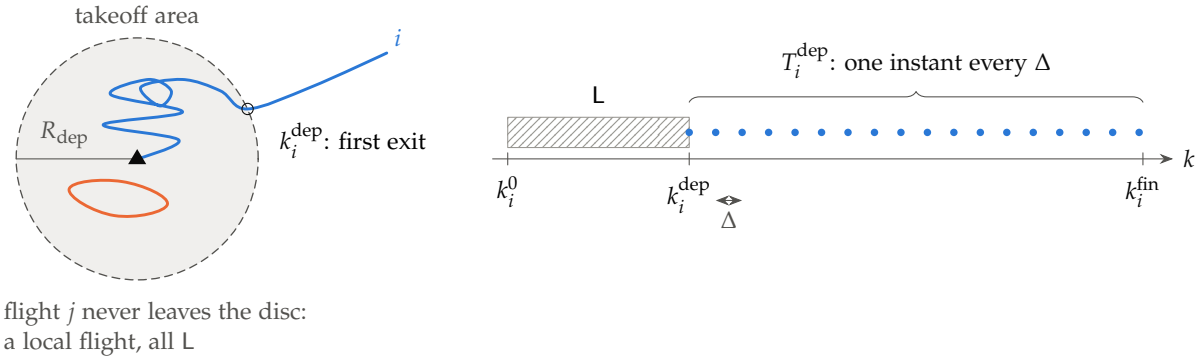


Figure 4: Local phase. Left: pilot i circles inside the disc of radius R_{dep} around the takeoff and departs at k_i^{dep}, its first fix outside; flight j never leaves and is local. Right: the same flight in time: L before the departure, then the grid instants T_i^{dep}.

3.2 Correspondences and contact

Definition 3.1 (correspondence)**.** For two flights $i \neq j$, the set of correspondences is

$$\begin{aligned}\mathcal{M}_{ij} = \Big\{(t,t') \in T_i^{\text{dep}} \times T_j^{\text{dep}} : |t'-t| \le \delta, \ \ &\|\mathbf{x}_i(t) - \mathbf{x}_j(t')\| < \varepsilon_h, \\ &|z_i(t) - z_j(t')| < \varepsilon_z, \ \ D_{ij}(t,t') = 1\Big\},\end{aligned} \tag{2}$$

where the heading condition is active only when both pilots glide:

$$D_{ij}(t,t') = \begin{cases} \mathbf{1}\big[\hat{\mathbf{v}}_i(t) \cdot \hat{\mathbf{v}}_j(t') > \cos\theta_{\max}\big] & \text{if } s_i(t) = s_j(t') = \text{transition}, \\ 1 & \text{otherwise.} \end{cases} \tag{3}$$

> **In words.** $\mathcal{M}_{ij}$ is a set of *pairs of instants* of the same clock. The pair (t,t') belongs to it when j, at instant t', is where i is at instant t: within ε_h horizontally and ε_z vertically, at most δ apart in time, and heading the same way if both glide. In the plane (t,t') it is a region of the band $|t'-t| \le \delta$ (Figure 5).

Definition 3.2 (contact)**.**

$$m_{ij}(t) = \mathbf{1}\big[\exists\, t' : \ (t,t') \in \mathcal{M}_{ij}\big], \quad m_{ji}(t') = \mathbf{1}\big[\exists\, t : \ (t,t') \in \mathcal{M}_{ij}\big], \quad a_{ij}(t) = m_{ij}(t) \vee m_{ji}(t). \tag{4}$$

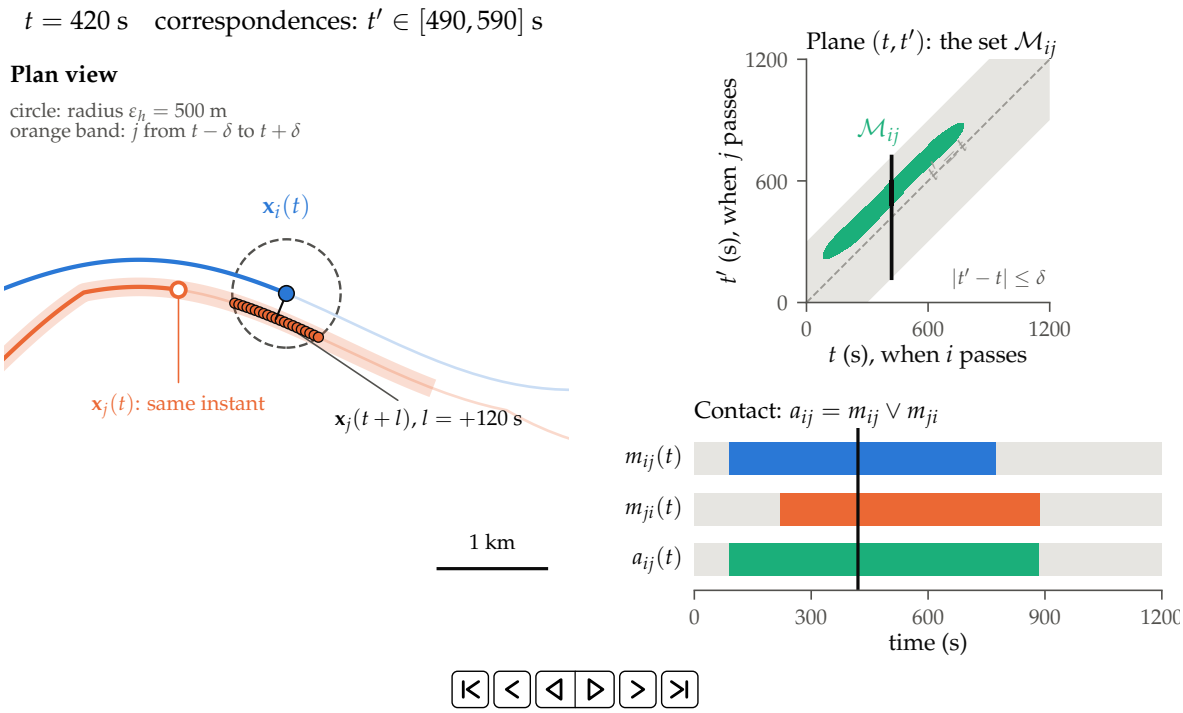


Figure 5: ▶ Correspondences and contact for a synthetic pair; the animation runs the instant t from 0 to 1200 s. Left: plan view around i at $t = 420$ s. Pilot j flies the route of i 120 s later and 250 m to the side; the circle has radius ε_h; the orange band is the track of j flown between $t - \delta$ and $t + \delta$; the dark dots are the instants t' with $(t, t') \in \mathcal{M}_{ij}$, and the segment joins $\mathbf{x}_i(t)$ to the closest of them, at delay $l = +120$ s. At the same instant j is about 1 km away. Top right: $\mathcal{M}_{ij}$ (green) in the band $|t' - t| \leq \delta$ (grey); the black segment is the column of the left panel. Bottom right: m_{ij} and m_{ji}, the shadows of $\mathcal{M}_{ij}$ on the two axes, and their union a_{ij}: i is on the path of j from 90 to 775 s, j on the path of i from 220 to 885 s. Model: i flies at 8 m/s along $y = 600 \sin(2\pi s/8\,\text{km})$ m, where s is the distance flown; j joins the route at 300 s and leaves it at 800 s, drifting sideways at 3 m/s; altitude and heading conditions hold throughout.

So $m_{ij}(t) = 1$ when i, at instant t, is at a place that j visits within δ, before or after, and $m_{ji}(t) = 1$ when j is at a place that i visits within δ. The contact $a_{ij}(t) = 1$ when at least one of the two is on the path of the other. The conditions of (2) are symmetric, so $\mathcal{M}_{ji}$ is $\mathcal{M}_{ij}$ with the coordinates swapped and $a_{ij} = a_{ji}$; the two halves m_{ij} and m_{ji} differ in general (Appendix E).

With the horizontal distance alone, $m_{ij}(t) = 1$ is the validity of instant t of Li et al. [16, §IV-A] (Appendix A.2).

3.3 The role of each term

Each term of (2) translates one of the requirements of Section 1.

Delay. With $t' = t$ the condition is simultaneous proximity; with $t' \neq t$ it accepts shifted tracks, as (R2)–(R3) ask. If $t' > t$, j passes where i was, later: j follows i, and $\mathcal{M}_{ij}$ lies above the diagonal (Figure 5). The contact asks only that some delay exists; its value is the subject of Appendix B.

Tolerances: a flat box. ε_h translates the "nearby places" of (R3). Height gets a tolerance of its own because a group is flat: horizontally it spreads over a kilometre, vertically it overlaps within a few hundred metres. A single radius, $\|\Delta\mathbf{x}\|_{3D} < \varepsilon$, either makes altitude irrelevant, if it is of the order of a kilometre, or penalises the normal horizontal spread in glide, if it is of the

order of 300 m (Figure 6). In a thermal the height is the only condition that acts: the pilots circle over one spot, with spiral radii of 20–26 m [1], far below ε_h, so only ε_z separates pilots circling at different heights. Since j climbs during the delay, the separation allowed there is ε_z plus the height j gains in δ: $|z_i(t) - z_j(t)| < \varepsilon_z + w\delta$, 540 m at a climb rate $w = 2$ m/s with $\delta = 2$ min.

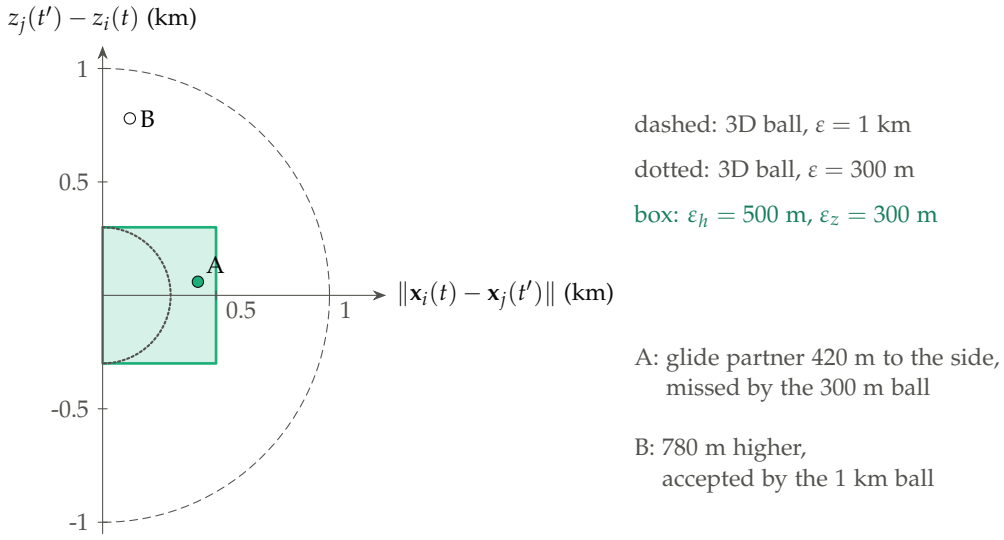


Figure 6: Tolerance regions for the separation between $(\mathbf{x}_i(t), z_i(t))$ and $(\mathbf{x}_j(t'), z_j(t'))$. The box of (2) is wide and flat, like a group: it keeps A and rejects B. A single 3D radius either rejects A or accepts B.

Heading only in glide. In glide the path is nearly straight and $\hat{\mathbf{v}}$ gives the direction of the route: condition (3) excludes a pilot who crosses the area heading elsewhere (Figure 7). In a climb the heading turns by 2π in about twenty seconds, and in a search the pilot turns often, with partial circles: in both the instantaneous heading says little about the route. So the condition is off whenever one of the two pilots is not in transition, unclassified fixes included. The flights without phases (Section 2) are grouped all the same: between two of them the condition never acts, and a correspondence rests on distance, height and delay alone. We rejected the velocity averaged over a few minutes with a minimum speed: in a long thermal without wind the mean velocity is nearly zero, and the threshold would remove the contacts exactly while two pilots share the thermal.

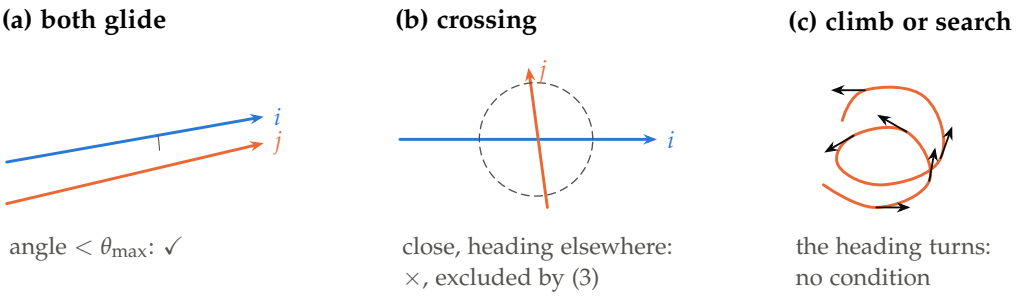


Figure 7: The heading condition (3). (a) Two gliders on the same route: their headings differ by less than $\theta_{\max}$. (b) A glider that crosses the area of i heading elsewhere is close, and the condition excludes it. (c) In a climb or a search the heading turns continuously and says nothing about the route: the condition is off.

3.4 Choice of the maximum delay

How large may the delay be? If j retraces the track of i with a constant delay ℓ, $\mathbf{x}_j(t) = \mathbf{x}_i(t-\ell)$, the two pilots are, at the same instant, as far apart as two points of the track of i flown ℓ apart,

$$\|\mathbf{x}_i(t) - \mathbf{x}_j(t)\| = \|\mathbf{x}_i(t) - \mathbf{x}_i(t-\ell)\| \le v_{\max}\,\ell,$$

and $v_{\mathrm{xc}}\,\ell$ on average once ℓ is longer than a climb–glide cycle, v_{xc} being the mean speed of advance. Asking that two pilots shifted by any delay up to δ stay within visual range gives $v_{\mathrm{xc}}\,\delta \le \varepsilon_{\mathrm{vis}}$ (Figure 8):

$$\delta \lesssim \frac{\varepsilon_{\mathrm{vis}}}{v_{\mathrm{xc}}}, \tag{5}$$

so $\delta \approx 5$ min for $\varepsilon_{\mathrm{vis}} \approx 1.5$ km and $v_{\mathrm{xc}} \approx 5$ m/s, with v_{xc} to be measured on the data, for example as the daily median of the speed of advance over 30-minute windows. The bound is the largest admissible delay. The delay at which following ends was measured on the data directly: 2 min (Section 6.1).

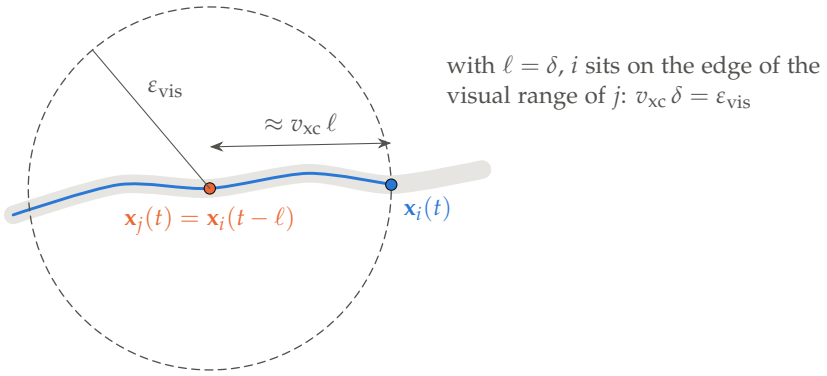


Figure 8: Choice of the maximum delay. A pilot that retraces the route of i with delay ℓ is about $v_{\mathrm{xc}}\ell$ behind; the dashed circle is the visual range around j. The largest delay that keeps i in sight is $\delta = \varepsilon_{\mathrm{vis}}/v_{\mathrm{xc}}$.

Remark 3.3 (two notions of "together"). Flying together may mean *seeing each other*, and so being able to use social information, or *exploiting the same air masses*:

$$\begin{aligned}
\text{visual co-presence:}\quad & V_{ij}(t) = \mathbf{1}\Big[\big\|\big(\mathbf{x}_i(t), z_i(t)\big) - \big(\mathbf{x}_j(t), z_j(t)\big)\big\| < \varepsilon_{\mathrm{vis}}\Big],\\
\text{shared air:}\quad & m_{ij}(t) = \mathbf{1}\big[\exists\, |l| \le \delta :\ \|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\| < \varepsilon_h,\\
& \qquad\qquad |z_i(t) - z_j(t+l)| < \varepsilon_z,\ D_{ij}(t, t+l) = 1\big].
\end{aligned}$$

Visual co-presence is simultaneous and depends on the 3D distance; shared air admits a delay and needs tight tolerances, vertical ones included. Definition 3.2 with ε_z of a few hundred metres is shared air, kept within visual range by (5); a wider ε_z moves it towards visual co-presence. So ε_z is a choice of definition, to be declared (Figure 9).

What we keep. A symmetric contact $a_{ij}(t)$ for every pair and every instant after the departures, with a delay up to δ, a flat box of tolerances and headings compared in glide. Over each stretch of contact a signed delay ℓ_{med} also says who follows whom; it does not enter the groups, and Appendix B defines it.

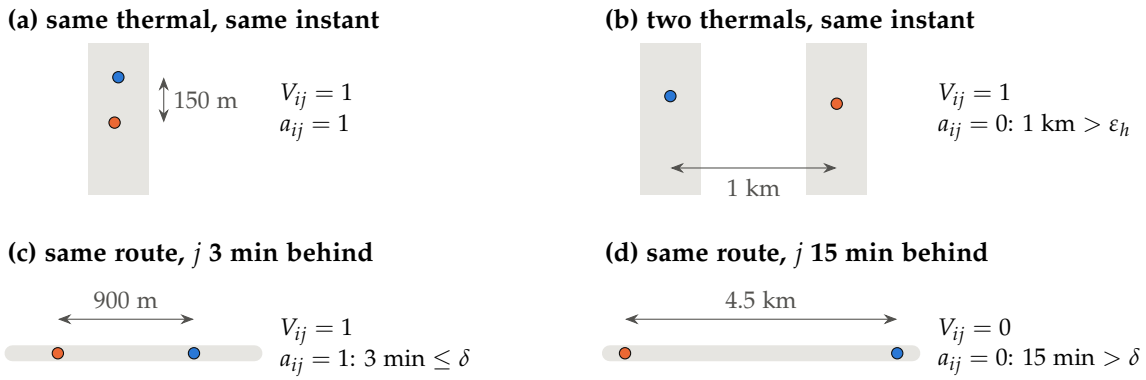


Figure 9: Visual co-presence V_{ij} and contact a_{ij} in four situations, with $\varepsilon_{\text{vis}} = 1.5$ km, $\delta = 5$ min and $v_{\text{xc}} = 5$ m/s. In (d) a delay of 15 minutes would put in contact two pilots who cannot see each other: the bound (5) prevents it.

4 From pairs to groups

Who flies with whom, as a group? The contact links two pilots at a time: it flickers, it is not transitive, and it changes during the afternoon. This section turns it into groups in four steps: pair intervals, configurations at each instant and groups over time, chains of groups (episodes), and the flights tied by them (dependency blocks).

4.1 Pair intervals

The instantaneous contact flickers wherever a quantity sits close to its threshold (a distance near ε_h, a height difference near ε_z, an angle near $\theta_{\max}$), at the changes of state, where the heading condition switches on and off, and where the phase labelling is wrong on a few fixes.

Definition 4.1 (pair interval)**.** For each pair consider the maximal runs of consecutive grid instants with $a_{ij} = 1$. Two runs separated by a gap shorter than g are joined; the resulting runs shorter than τ^* are dropped. Each remaining run is a half-open interval $[t_{\text{on}}, t_{\text{off}})$, and the set of the intervals of the day is $\mathcal{I}$.

Joining the short gaps plays the role of the robustness of Buchin et al. [7, §4], which ignores interruptions shorter than α; the maximal segments of Li et al. [16, 22] are an alternative. The minimum duration τ^* keeps chance contacts out, and the null model fixes it (Section 5).

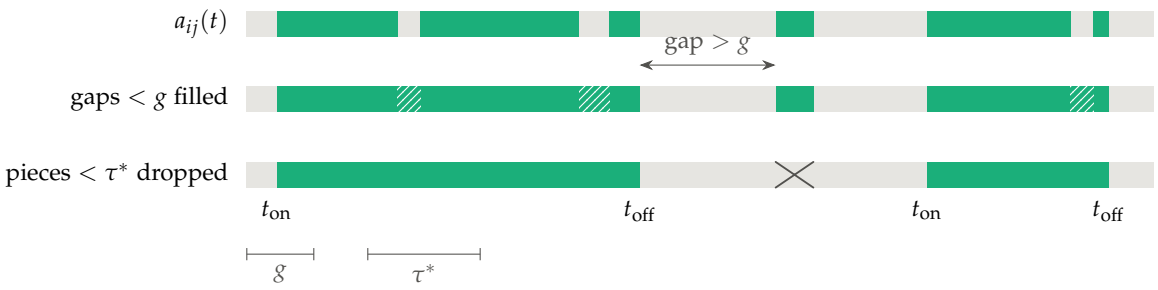


Figure 10: From the contact to the pair intervals. Top: the contact of one pair, with the flicker of the instantaneous condition. Middle: gaps shorter than g are filled (hatched). Bottom: runs shorter than τ^* are dropped; two intervals remain.

4.2 The problem of non-transitivity

If 1 flies with 8 early in the afternoon and 8 with 14 later, 1 and 14 may never have met: contact is symmetric but not transitive. A partition into groups needs an *equivalence relation* $\sim$, reflexive ($i \sim i$), symmetric ($i \sim j \Rightarrow j \sim i$) and transitive ($i \sim j$ and $j \sim k \Rightarrow i \sim k$), so any definition of groups must give up something (Figure 11):

(a) direct contact between all members: the transitive closure puts 1 and 14 in one group although they never met;

(b) the partition: groups may overlap, here $\{1, 8\}$ and $\{8, 14\}$, as the maximal groups of Buchin et al. [7] do;

(c) the static picture: the groups are defined instant by instant.

We take (c): at every instant the configurations are the connected components of the graph of active contacts, as the components of [7]; a group lasts only as long as its members stay one component, and the episodes describe the evolution without giving a group a lasting identity.

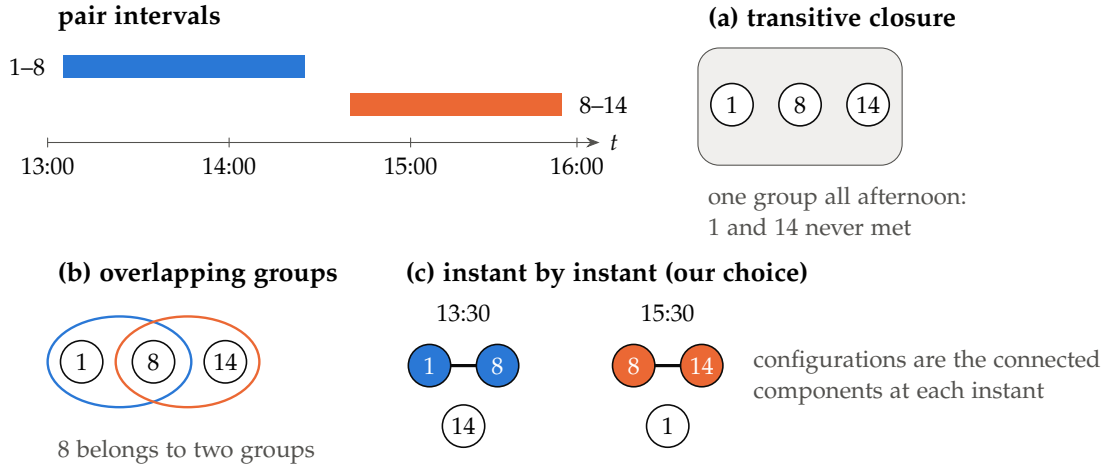


Figure 11: Non-transitivity: 1–8 early in the afternoon, 8–14 later, and 1 and 14 never meet. The three ways out: (a) one group of all three, (b) two overlapping groups, (c) the groups of each instant, our choice.

4.3 Configurations and groups

The active graph at instant t is

$$\begin{aligned} G(t) &= \big(V(t), E(t)\big), \qquad V(t) = \{i : \ t \in T_i^{\text{dep}}\}, \\ E(t) &= \big\{\{i,j\} : \ \exists\,(i,j,t_{\text{on}},t_{\text{off}}) \in \mathcal{I},\ t \in [t_{\text{on}},t_{\text{off}})\big\} : \end{aligned} \tag{6}$$

its nodes are the pilots that at t have departed, have not landed and have a position; its edges are the pairs whose interval contains t. A connected component of $G(t)$ with at least two nodes, with its edges, is a *configuration* at instant t: its pilots are joined by chains of contacts, and its edges say who is in contact with whom (Appendix F recalls the terms).

Definition 4.2 (group)**.** A group g is given by a set of flights M_g, its members, with $|M_g| \geq 2$, and by an interval of time $[t_g^{\text{start}}, t_g^{\text{end}})$, such that M_g is the set of nodes of a configuration of $G(t)$ at every t of the interval and the interval is maximal. We write $g = \big(M_g, [t_g^{\text{start}}, t_g^{\text{end}})\big)$.

The members of a group are fixed: a pilot who joins or leaves ends it and starts another. Only the edges among the members may change, and each change gives a new configuration of the same group: an edge that closes a triangle turns a chain into a triangle, and the group goes on (Figure 12). Over time, a configuration keeps its edges over a maximal interval, and the configurations of a group follow one another without gaps over the interval of the group. The groups are the edges of the Reeb graph of [7] with at least two entities (Figure 29), built on delayed contacts instead of simultaneous proximity.

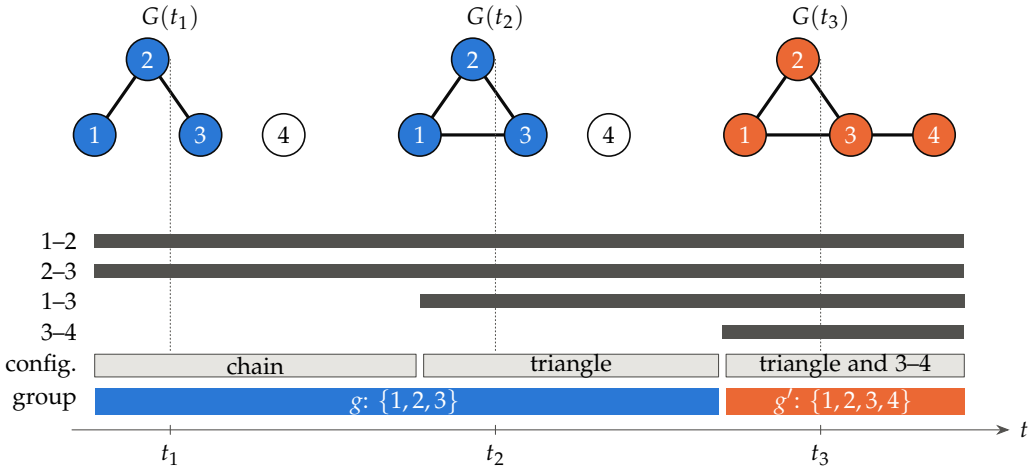


Figure 12: Configurations and groups. Top: the active graph at three instants. Bottom: the pair intervals of the four flights, the configurations and the groups they produce. At t_2 the edge 1–3 closes a triangle: a new configuration with the same members, so g goes on. When 4 joins, g ends and g' starts.

Proposition 4.3. *At every instant each flight belongs to at most one configuration, hence to at most one group.*

Proof. The connected components of $G(t)$ form a partition of $V(t)$. □

The membership of flight i at instant t is therefore well defined:

$$\gamma_i(t) = \begin{cases} \mathsf{L} & \text{if } k_i^0 \le t < k_i^{\text{dep}} \text{ (local phase), or the flight is local,} \\ g & \text{if } i \in M_g \text{ and } t \in [t_g^{\text{start}}, t_g^{\text{end}}), \\ \mathsf{S} & \text{if } t \in T_i^{\text{dep}} \text{ otherwise (solo flight).} \end{cases} \tag{7}$$

In the tables, every instant between the departure and the landing outside a group is labelled S, the instants in a gap of the track included, where the flight has no position. The analyses that read the fixes see nothing there; only the time fractions computed from the membership include them.

4.4 Episodes

A group that gains or loses a pilot becomes a new group, yet its pilots go on flying together. Episodes keep this continuity.

Definition 4.4 (episode)**.** Two groups are linked, $g \to g'$, if one starts exactly when the other ends and they have at least one member in common:

$$g \to g' \iff t_{g'}^{\text{start}} = t_g^{\text{end}} \ \wedge \ M_g \cap M_{g'} \neq \emptyset. \tag{8}$$

An episode is a connected component of the undirected graph associated with $\to$: the groups tied together by links, directly or through others.

So an episode is a chain of groups without interruption: each starts when the previous one ends and keeps at least one of its members, and two groups of the same episode need not share anyone. In the language of [7], it is a connected component of the Reeb graph without the edges of single pilots. Four properties matter (Figure 13):

(i) a split produces two groups linked to the same previous one, so they stay in the same episode; the same holds for a merge;

(ii) if the group dissolves completely, the chain breaks and the episode ends; a group that forms again later opens a new episode;

(iii) no threshold is needed: one member in common is enough;

(iv) the same flight can belong to several episodes of the same day, at different times.

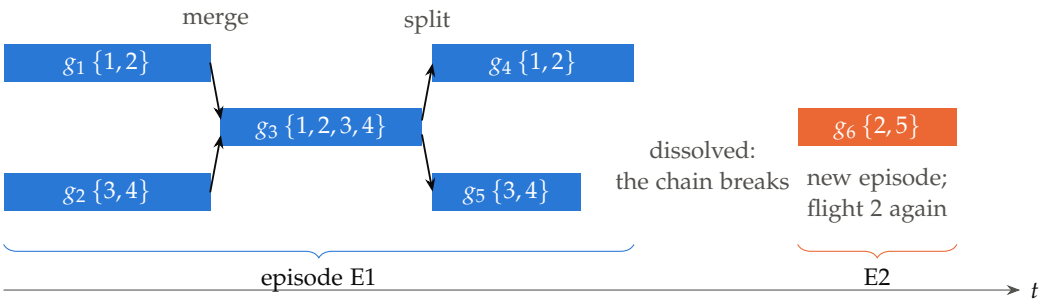


Figure 13: Episodes. Arrows are the links (8). The merge of g_1 and g_2 into g_3 and the split of g_3 into g_4 and g_5 keep all five groups in episode E1, although g_1 and g_5 share no member. After the group dissolves, g_6 has no predecessor and opens E2: flight 2 belongs to both episodes.

Remark 4.5 (why not a persistent group name). One could keep the same group name while the overlap of the members between two instants exceeds a threshold, as methods that track communities in evolving networks do with the Jaccard index [10]. The choice has three defects: the threshold is arbitrary; a group whose members are replaced one at a time keeps its name even when none of the first members is left; in splits and merges no natural criterion says who inherits the name. So we let the episode record only the continuity, and the group the composition.

4.5 Dependency blocks

Which flights can the statistics treat as independent? Those that never flew in the same episode, directly or through a chain of episodes.

Definition 4.6 (block)**.** For each day consider the bipartite graph flights–episodes, with an edge between a flight and every episode it belongs to. Its connected components, restricted to the flights, are the *dependency blocks*; a flight never in a group is a block on its own.

Proposition 4.7. *The blocks are the finest partition of the flights in which the members of every episode are in the same block.*

Proof. A partition with this property must contain the transitive closure of the relation "share an episode", whose classes are exactly the connected components of the bipartite graph; these, in turn, satisfy the property. □

The blocks are the units resampled in the bootstrap (Section 8.1): flights of different blocks never shared an episode. Buchin et al. already note that a grouping structure splits the trajectories into independent data sets [7, §1].

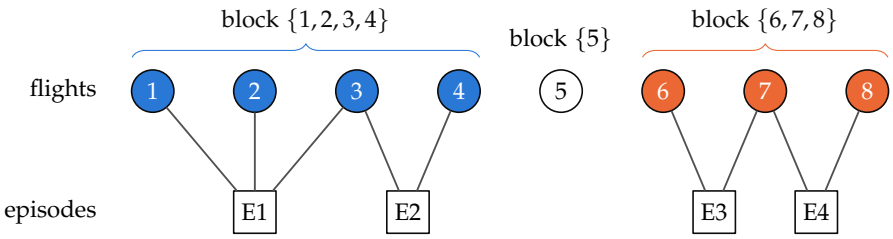


Figure 14: Dependency blocks of a day. Flight 3 is in episodes E1 and E2, so flights 1–4 form one block although 1 and 4 share no episode; flight 5 never flies in a group.

4.6 An example day

Figure 15 follows a day with five flights. At 11:40 flights 1, 8 and 22 meet (edges 1–8 and 8–22): group $g_1 = \{1, 8, 22\}$ starts, in the configuration of a chain. At 12:05 the edge 1–22 closes the triangle: a new configuration with the same members, so g_1 goes on; at 12:30 the edge closes and the chain is back. At 12:55 the interval 1–8 ends as 1 changes route: g_1 ends and $g_2 = \{8, 22\}$ starts; they share 8 and 22, so $g_1 \to g_2$. At 13:30 the edge 8–14 opens: $g_3 = \{8, 14, 22\}$, and $g_2 \to g_3$. At 14:10 the group dissolves and episode E1 ends. At 15:00 flights 8 and 22 meet again: $g_4 = \{8, 22\}$ has no predecessor and opens episode E2.

So flights 1 and 14 share an episode without ever sharing a group; 8 and 22 belong to two episodes; 1, 8, 14 and 22 form one dependency block, and flight 5, never in a group, a block of its own.

What we keep. At every instant each flight has one label, L, S or its group; the configurations record how the members of a group are linked, groups chain into episodes, and episodes tie the flights into blocks. These labels are the output of the method (Appendix D). They still depend on τ^*, which Section 5 fixes.

5 Calibration: what chance means

How many contacts would chance alone give? Pilots who never interact still meet: they take off from the same site, follow the same ridges and climb in the same house thermals. A contact says little by itself; what counts is how many contacts of a given duration occur beyond chance. Interaction indices based on standard tests overestimate the interactions with finely sampled telemetry [17], so we build chance from the data, with copies of the flights shifted in time, and derive from it the shortest duration τ^* of a contact. The run reported from here on covers the 162 485 cleaned flights of 6 209 dates of the FFVL archive (Section 2).

5.1 The null model

On each date we draw up to 200 pairs (i, j) of flights whose takeoffs are less than $R_s = 10$ km apart, and replace flight j by its copy shifted in time:

$$\mathbf{x}_j^s(t) = \mathbf{x}_j(t + r_{ij}), \quad z_j^s(t) = z_j(t + r_{ij}), \qquad r_{ij} = (k_j^0 - k_i^0) - \vartheta, \quad \vartheta \sim \hat{P}(k_j^0 - k_i^0). \tag{9}$$

The delay ϑ is drawn from the delays between the takeoffs of the pairs of that date with takeoffs within R_s: the copy of j takes off ϑ after i, as a real pilot of the day could. We accept only shifts with $|r_{ij}| \geq r_{\min} = 2\delta + \Delta = 245$ s, rounded to the grid step, and a pair enters once, with the roles of i and j drawn at random. The idea of shifting tracks in time, which keeps the spatial

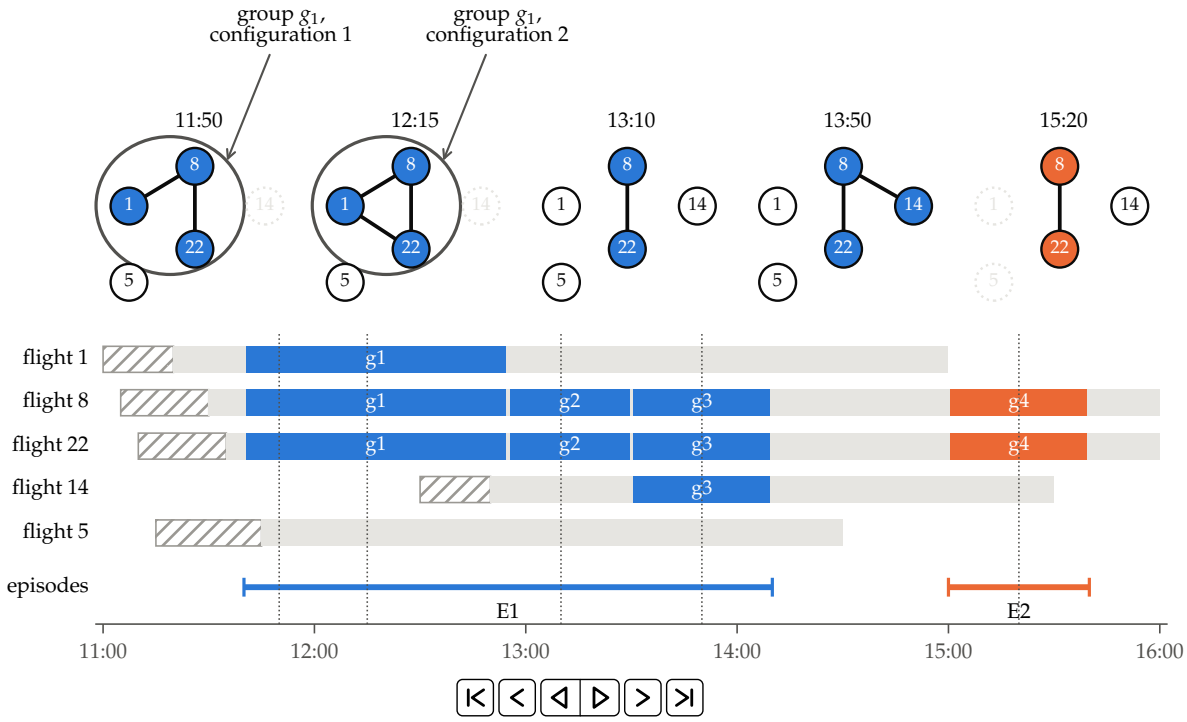


Figure 15: ▶ An example day; the animation runs the day from 11:00 to 16:00 and stops at each event. Top: the active graph $G(t)$ at the five dotted instants; filled nodes are in a group, coloured by episode; dashed nodes are in the local phase, faint ones not in flight. Bottom: the membership $\gamma_i(t)$ of each flight: hatched is the local phase L, grey is solo S, coloured segments are the groups g_1–g_4, and the last row spans the episodes.

constraints and destroys the synchronisation, comes from Spiegel et al. [24], who shuffle whole days of each individual; here a flight is shifted within its day, against one other flight.

No real correspondence survives the shift (Figure 16). If (t, t') was a real correspondence, in the surrogate pair the position of j at t' comes at $t' - r_{ij}$, at least $|r_{ij}| - \delta > \delta$ away from t. The correspondences that remain are encounters on the same route without synchrony.

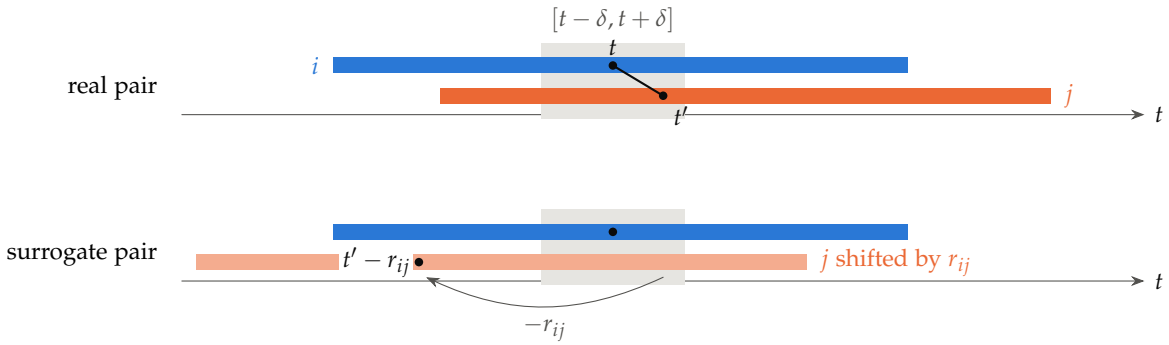


Figure 16: Null model. Top: a real correspondence (t, t'), with t' inside the window $[t-\delta, t+\delta]$ (grey). Bottom: j shifted by r_{ij}, $|r_{ij}| > 2\delta$; the same point of j now comes at $t' - r_{ij}$, outside the window.

The shifted pair keeps the route, the terrain, the date, and the takeoff delays of the day as a distribution. On the archive the shift is far larger than its minimum: 58 min in the median, and 90% of the shifts lie between 8 min and 4.4 h (Figure 43 in Appendix H). So the copy also flies in the weather of another hour, with other thermals working and another cloud base.

Chance is therefore an operational notion: the contacts that the same routes give without

synchrony. Two consequences follow.

- Pilots who use the same thermals at the same time without following each other share the weather of the hour, which the shift removes: their contacts count as real.
- A pilot who follows another deliberately, but more than δ behind, never makes a contact in the real pair, while a surrogate whose shift brings that delay within δ counts it as chance.

5.2 The false discovery rate and τ^*

The surrogate pairs go through Phases 1–3 like the real ones (Appendix C.1); their pair intervals are the contacts chance alone gives. They are a sample of the pairs of each date, so we count chance per second of exposure, date by date, and scale it to the real pairs. On date d:

- $\mathcal{P}_d$ is the set of the pairs (a, b), $a < b$, of flights of the date with takeoffs less than R_s apart; their exposure is $H_d = \sum_{\mathcal{P}_d} h_{ab}$, with $h_{ab} = \max\big(\min(k_a^{\text{fin}}, k_b^{\text{fin}}) - \max(k_a^{\text{dep}}, k_b^{\text{dep}}),\, 0\big)$ the time both fly after departure;
- S_d is the same sum over the surrogate pairs (i, j, r) of the date, with the times of j moved by $-r$;
- $\rho_{0,d}(\tau) = M_d(\tau)/S_d$ is the number of chance intervals per second of exposure, with $M_d(\tau)$ the intervals of the surrogate pairs lasting at least τ.

Over the archive, with $N(\tau)$ the number of real intervals of the pairs of every $\mathcal{P}_d$ lasting at least τ:

$$C(\tau) = \sum_d \rho_{0,d}(\tau)\, H_d, \qquad \widehat{\text{FDR}}(\tau) = \frac{C(\tau)}{N(\tau)}, \qquad \tau^* = \min\big\{\tau :\ \widehat{\text{FDR}}(\tau) \le q\big\}. \tag{10}$$

Dates without surrogates add to N only, and τ runs over the grid $\{0, 5, \ldots, 14\,400\}$ s.

> **In words.** $C(\tau)$ is the number of intervals lasting at least τ that chance alone would give the real pairs. Dividing it by the number actually observed gives the share of chance intervals among them; τ^* is the shortest duration at which that share falls to q, one in ten.

In analogy with SAM [28], this is the number of discoveries expected by chance over the number observed, with time shifts in place of the permutations.

Most contacts are chance (Figure 17): 74% of all pair intervals with $\delta = 2$ min (77% with 5 min), 47% of those lasting 10 min, 16% of those lasting 30 min. Short contacts say little; only long ones are mostly real. So τ^* is tens of minutes: it is the duration at which the real contacts outnumber chance by the factor $1/q$. Resampling the dates 200 times moves τ^* by a few minutes (Table 6): the curve sets it, and the noise of the surrogates barely moves it.

5.3 How much the shift matters

Does the length of the shift change what counts as chance? We recomputed the false discovery rate with only the surrogate pairs of each date whose shift falls in one class, each class with its own exposure S_d and intervals, on the 2 737 dates (of 6 209) that have surrogates in every class (Table 3).

It does, by more than a factor two. Short shifts give 2.8 times the chance of long ones at 10 min and 5.7 times at 30 min, and τ^* goes from 23 min with shifts of an hour or more to

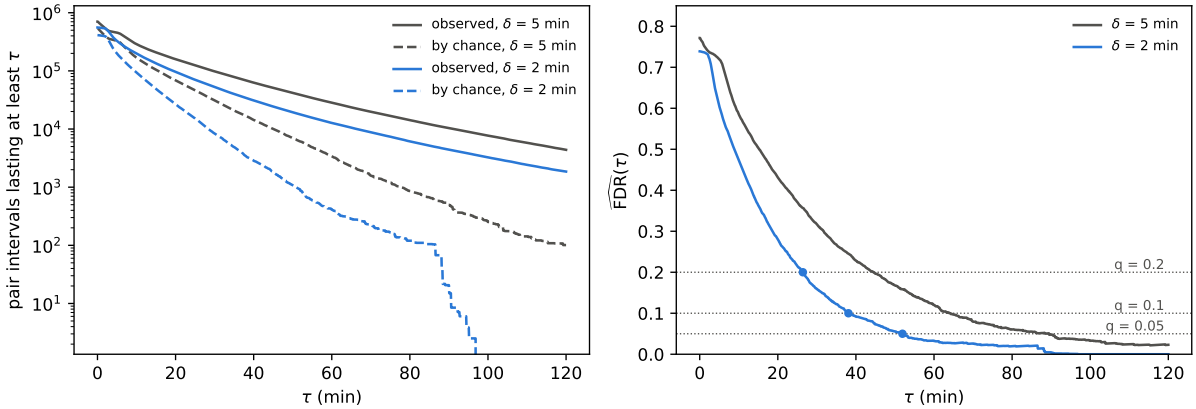


Figure 17: Left: real pair intervals lasting at least τ (solid) and those chance alone gives (dashed), with $\delta = 5$ min (grey) and $\delta = 2$ min (blue), on the whole archive. Right: their ratio, the false discovery rate of Eq. (10); dots mark τ^* for $q = 0.05$, 0.1 and 0.2 with $\delta = 2$ min.

Surrogates	pairs	$\widehat{\text{FDR}}$(10 min)	$\widehat{\text{FDR}}$(30 min)	τ^* (min)	5–95% over dates
all shifts	241 824	0.48	0.16	38.2	37.0–41.5
$\lvert r\rvert < 30$ min	69 444	0.73	0.31	53.3	49.7–72.0
$\lvert r\rvert \geq 60$ min	119 687	0.26	0.05	23.0	21.1–24.4

Table 3: The false discovery rate and τ^* ($q = 0.1$, $\delta = 2$ min) with the surrogates of one class of shifts, on the 2 737 dates with surrogates in each class.

53 min with shifts under 30 min; with all the shifts it is 38 min, as on the whole archive. Two effects add up and cannot be separated here. Short shifts keep the weather of the hour, and they also bring within δ the pilots who follow each other a few minutes apart, who are many: real correspondences are still 2 to 3 times chance between 195 and 500 s (Section 6.1).

What we keep. Chance is what the same routes give without synchrony, with shifts that follow the takeoff delays of the day. A contact counts only if it lasts at least τ^*, the duration at which at most a share q of the contacts is chance; q and δ are chosen in Section 6.

6 The parameters on the data

Which values should the parameters take, and which of them can the data decide? For each one we ask a question, compare real pairs with chance (Section 5.1) and choose; the profiles use a sample of 300 random dates, the choices of δ and q the whole archive. Table 4 sorts the parameters in three kinds. The *tolerances* δ, ε_h, ε_z, $\theta_{\max}$ and g say when two positions match. The *duration* τ^* says how long a match must last, and follows from the share q of chance contacts we accept (Section 5.2). The *fixed settings*, in the lower part of the table, make the computation exact or fast and are not tuned.

6.1 The maximum delay δ

How far behind can a pilot be and still fly with the other? The visual range bounds δ at about 5 min (Section 3.4); here the data say how long the delay between two pilots who fly together actually is.

Parameter	Value	How it was set
δ, largest delay	2 min	measured on the delay profile, Section 6.1
ε_h, ε_z	500 m, 300 m	definitions: profiles without an edge, Section 6.2
g, longest gap joined	120 s	definition: gaps of real and chance contacts alike, Section 6.2
$\theta_{\max}$	45°	no effect on the result, Section 6.3
q, accepted false discovery rate	0.1	trade-off, Section 6.4
τ^*, shortest contact	38.1 min	computed from q and the null model, Section 5.2
Δ, grid step	5 s	a pilot at 100 km/h is at most 69 m from a grid position, against $\varepsilon_h = 500$ m (Appendix C.2)
$h_{\max}$	10 s	positions are interpolated only between fixes at most 10 s apart; the flights of coarser loggers are left out of the statistics
R_{dep}	3 km	contacts around the takeoff, before the first exit from this disc, are not counted (Section 3.1)
$v_{\max}$	100 km/h	speed of the candidate filter: pieces of track end at every faster step, so the filter never misses a correspondence (Lemma C.1, checked against every pair of instants on synthetic days); the profiles of Section 6.2 widen it to 3 km, the largest distance they count
R_s, $r_{\min}$	10 km, $2\delta + \Delta$	the null model, with at most 200 surrogate pairs per date (Section 5.1)

Table 4: The parameters and how each was set. Top: chosen on the data or declared as definitions. Bottom: fixed settings.

The delay profile. For every delay l we counted the correspondences $(t, t + l)$ of the real pairs and of the surrogate pairs, per second of exposure, with a window of 15 min on 300 random dates (58 910 h of real exposure), and took their ratio:

$$R(l) = \frac{\text{real correspondences at delay } l \text{ per second of exposure}}{\text{chance correspondences at delay } l \text{ per second of exposure}}. \tag{11}$$

Chance is flat in the delay, as expected: a shifted copy has no preferred delay. The real pairs peak at a few seconds, where they have 5.6 times the correspondences of chance, and decay: R falls below 4 at 105 s, below 3 at 195 s and below 2 at 500 s, and is still 1.5 at 15 min (Figure 18). The excess has two parts. The fast one, within 2–3 min, is pilots flying together. The slow one, which has not ended at 15 min, is pilots who use the same thermals at about the same time without following each other, a synchrony that the shifts of the surrogates remove. A rule such as "the delay that holds 95% of the excess" would pick 13 min, because it counts the slow tail; so we read the ratio instead.

The delay inside the contacts. We computed $|\ell_{\text{med}}|$ (Appendix B) for every pair interval of the run with $\delta = 5$ min, whose window is wide enough to show where the delays fall: 965 770 intervals, of which 169 686 have all their instants in search and no delay. The longer the contact, the shorter the delay (Table 5; the whole distribution is in Appendix H). A pair that only shares route and weather matches at any delay of the window, and its closest correspondence is often at the largest ones: short intervals pile up at the edge. The contacts of at least 90 min, all above τ^* of this run (89.5 min), have a median delay of 25 s, and 90% of them lie within 2 min. Long contacts are pilots a few tens of seconds apart on the same path.

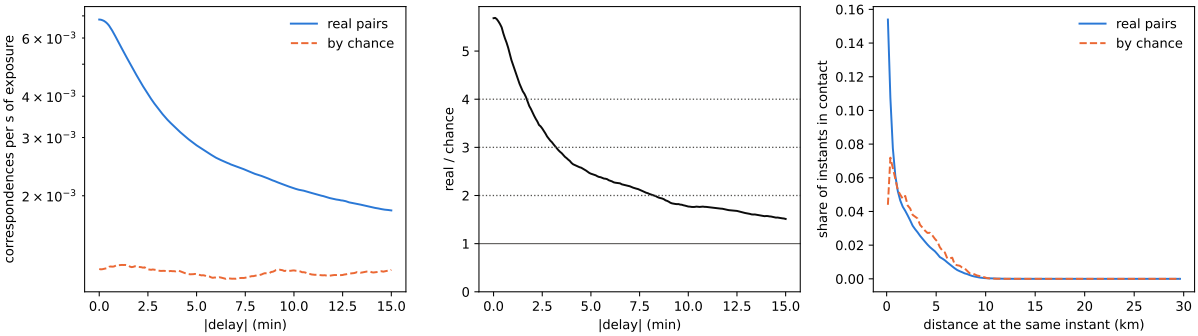


Figure 18: The delay profile on 300 dates with a 15 min window. Left: correspondences per second of exposure at each delay, real and by chance. Middle: their ratio $R(l)$ of Eq. (11), with the levels 4, 3 and 2. Right: the distance between the two pilots at the same instant when one is on the path of the other.

Duration of the interval	intervals	median $\lvert\ell_{\text{med}}\rvert$	within 2 min	at the edge
under 10 min	436 444	215 s	31%	30%
10–30 min	246 548	165 s	38%	14%
30–90 min	102 358	82.5 s	64%	2.4%
90 min or more	10 734	25 s	90%	7 intervals

Table 5: The delay of the pair intervals by their duration ($\delta = 5$ min, whole archive). At the edge: $\lvert\ell_{\text{med}}\rvert \geq 285$ s, the last 15 s of the window.

Choice. We take $\delta = 2$ min: within it the real correspondences are at least 3.7 times chance, and it contains nine in ten of the long contacts. A shorter delay also shortens τ^*, since fewer chance matches enter: from 89.5 to 51.9 min at $q = 0.05$ (Table 6).

The delay also bounds the distance between two pilots in contact at the same instant: at most $\varepsilon_h + v_{\max}\delta \approx 0.5 + 27.8 \times 120/1000 \approx 3.8$ km, below the 6 km we would still call a group; a group, a chain of pairs, can be wider. Even with the 15 min window, 58% of the instants in contact have the two pilots within 2 km and 94% within 6 km (Figure 18, right).

6.2 The tolerances against chance

Is there a distance, a height difference or a gap beyond which real pairs look like chance? Such an edge would give the natural value of ε_h, ε_z or g. In analogy with the delay profile, with $\delta = 2$ min and the null model of the chosen run, on the 210 of the 300 dates that have surrogates, we measured at every instant t of pilot i the smallest horizontal distance between i at t and j at $t + l$, $\lvert l\rvert \leq \delta$, over the delays that pass the vertical and heading conditions; the smallest height difference, over the delays that pass the horizontal and heading conditions; and the gaps between consecutive instants of contact of a pair.

None of the three has an edge (Figure 19). The horizontal ratio is 6.3 within 25 m, below 2 from 250 m, below 1.5 from 500 m, and still 1.13 between 2.5 and 3 km. The vertical ratio is 4.0 within 25 m and below 2 from 150 m; from 275 m on it stays near 1.5 (1.51 on average up to 1 km), the excess that the horizontal condition alone gives. Both values in use sit where real pairs are about 1.5 times chance.

Real pairs have 1.46 times the gaps of chance per hour, because they have more contact, but the lengths of the gaps are the same: the shares of each class differ by at most 2 points. The gaps come from how pilots fly in and out of each other's path, the same in real and in chance contacts. The distance between the two pilots at the same instant, counted without requiring a correspondence, has no edge either (Appendix H).

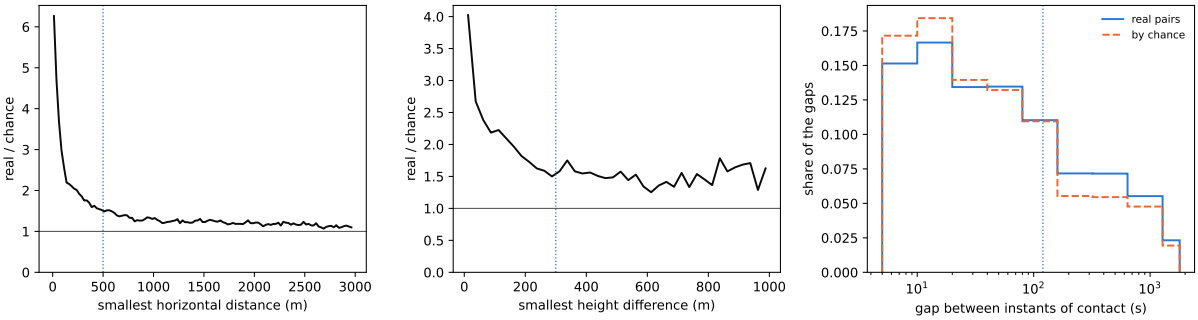


Figure 19: The tolerance profiles on 210 dates. Left and middle: real pairs over chance, per second of exposure, by the smallest horizontal distance and the smallest height difference to the path of the other pilot within δ; dotted, the values in use. Right: the lengths of the gaps between instants of contact, as a share of all gaps (the 8% of real and 9% of chance gaps longer than 30 min are not drawn); dotted, g.

Choice. The data single out none of ε_h, ε_z and g: we keep 500 m, 300 m and 120 s and declare them as definitions.

6.3 One parameter at a time

Does the result depend on the exact thresholds? A plateau is a range of values over which the result does not change; a value inside it makes the result insensitive to the exact threshold. We measure the result by f_G, the fraction of the flight time after departure spent in a group:

$$f_G = \frac{\sum_i \left|\{t \in T_i^{\text{dep}} : \gamma_i(t) \notin \{\mathsf{L}, \mathsf{S}\}\}\right|}{\sum_i |T_i^{\text{dep}}|}. \tag{12}$$

We changed one parameter at a time on 300 random dates, recalibrating τ^* each time, with $\delta = 5$ min (Table 15 in Appendix H). $\theta_{\max}$ changes nothing: contacts in glide are few. The others move f_G by up to a factor three, mostly through τ^*. There is no plateau, in agreement with the profiles: the thresholds decide the result, and the thesis must say so.

The check is coarse. On 300 dates τ^* is noisy (126 min with the starting values against 89.5 on the whole archive, and six settings stop at the same surrogate interval); a conclusive check needs the whole archive for each setting, about one hour each. It was not repeated with $\delta = 2$ min.

6.4 The false discovery rate q

How much chance do we accept among the contacts we keep? A smaller q keeps only very long contacts and few groups, almost all of them real; a larger q sees more groups, with more chance among them. Table 6 and Figure 20 give the trade-off.

δ	q	τ^* (min)	5–95% over dates	para in a group	hang in a group	episodes
5 min	0.05	89.5	79.5–91.5	8.9%	2.1%	6 090
2 min	0.05	51.9	49.1–53.3	13.7%	3.8%	10 553
2 min	0.1	38.1	37.0–41.1	21.7%	6.9%	17 169
2 min	0.2	26.4	25.2–27.2	31.7%	13.9%	26 504

Table 6: τ^* and the groups it gives. Flights with a logger of at most 10 s and a position after departure: 154 598 paragliders and 6 039 hang gliders.

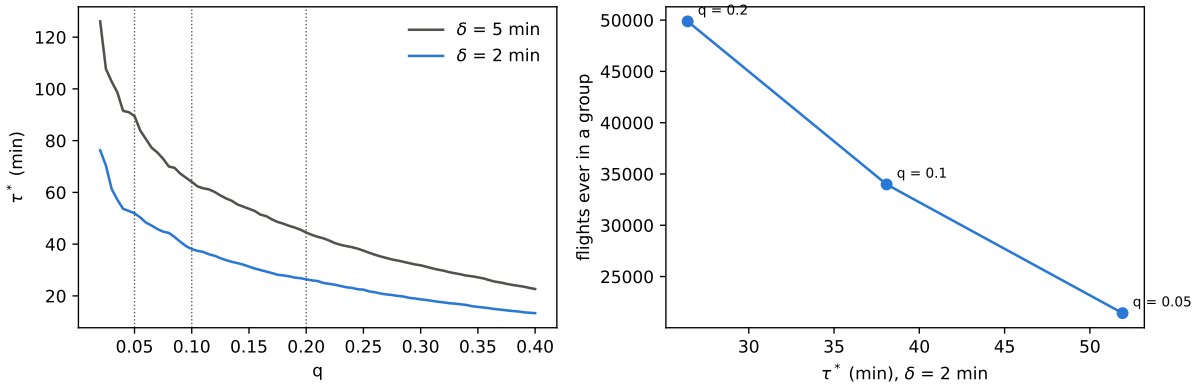


Figure 20: Left: τ^* as a function of q, for the two delays. Right: flights ever in a group against τ^*, for $q = 0.05$, 0.1 and 0.2 with $\delta = 2$ min.

Choice. We take $q = 0.1$: at most one contact in ten is chance, and $\tau^* = 38.1$ min. Going to 0.2 would lower τ^* to 26 min but would let chance make one contact in five.

What we keep. $\delta = 2$ min and $q = 0.1$ are our choices on the data, and give $\tau^* = 38.1$ min; ε_h, ε_z and g are definitions, and $\theta_{\max}$ does not matter.

7 What we find

What do the groups look like on the archive, and what holds them together? With the chosen values, $\delta = 2$ min and $q = 0.1$, 21.7% of the paraglider flights and 6.9% of the hang-glider flights fly in a group at some point, in 17 169 episodes (Table 6). Chance is again the null model of Section 5.1; bands are 5–95% of 200 draws of the dependency blocks, or of the dates where the captions say so. Graph terms are defined in Appendix F, the two models in Appendix G. The report *Group and solo flights* (`analysis/groups/group-identification`) compares the motion of pilots in a group and alone; some results here refer to it.

7.1 What the groups are

How large are the groups, how long do they last, and how do they change? We count episodes and groups (Section 4) by number and by duration; an episode lasts from the start of its first group to the end of its last, and its participants are the distinct flights it holds.

	q	number	median duration	2 participants items	2 participants time	5 or more items	5 or more time	largest
episodes	0.05	10 553	74.6 min	77%	67%	3.9%	8.1%	21
	0.1	17 169	58.2 min	74%	61%	5.7%	12.6%	53
	0.2	26 504	44.5 min	73%	56%	7.0%	17.2%	91
groups	0.05	20 275	52.7 min	70%	81%	5.6%	2.3%	18
	0.1	40 051	30.4 min	63%	77%	9.4%	3.5%	27
	0.2	75 953	17.1 min	56%	74%	14.1%	5.0%	40

Table 7: Episodes and groups for three values of q ($\delta = 2$ min). Items: share of the episodes or groups of that size; time: share of their total duration.

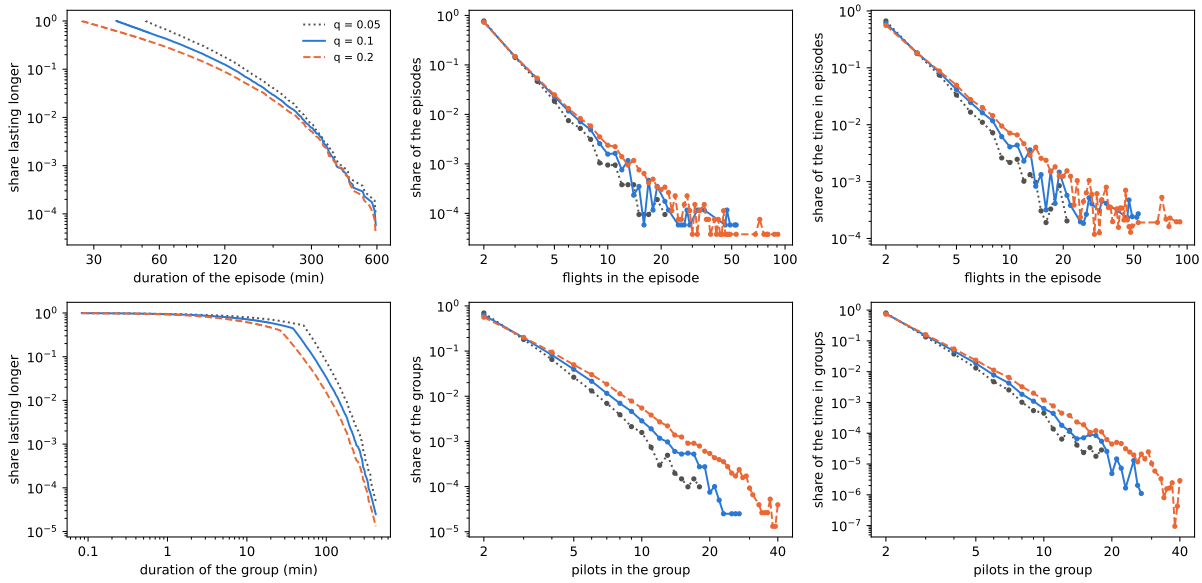


Figure 21: Episodes (top) and groups (bottom), $\delta = 2$ min, whole archive, for three values of q. Left: share lasting longer than a duration. Middle: share of each size. Right: share of the total duration in each size.

Sizes and durations. With the chosen values the median episode lasts 58.2 min and three in four are pairs (Table 7). Episodes last at least τ^*, since a contact must, and larger episodes last longer, so they weigh more in time than in number. Groups have no minimum duration: every arrival or departure of a member starts a new one, so 10% last less than 2.1 min, and the median is 30.4 min. Large groups change quickly, so in time the pairs weigh more than in number. 0.9% of the groups hold a hang glider. Inside a group the edges among the members can change: the 40 051 groups of the chosen values hold 52 856 configurations. A larger q, hence a shorter τ^*, gives more, shorter and larger groups: the shares of the small sizes change little, and the tail moves.

How groups change. A group ends in one of five ways: a *join* (members added), a *leave* (members removed in flight), a *landing* (a removed member lands within 5 s, the grid step), a *split* or a *merge* (Appendix F); a group of two pilots that loses one dissolves. Table 8 gives, by size, the groups ending per hour spent at that size.

- **Most episodes never change:** 74% of the 17 169 episodes are one group; 5.0% hold a split, 4.4% a merge.
- **Episodes last; groups change.** 12% of the episodes last 2 h or more. A group lasts 41 min with 2 pilots and 4 min with 8 or more: joins, leaves, splits and merges all grow with size, so a large group changes members every few minutes.
- **Groups break in flight:** 91% of the 18 376 groups that dissolve do so in flight, 9% at a landing; a pair ends mostly because the two separate. 34% of the 18 218 groups with no predecessor form when a member departs: pilots who took off together.

Where in the flight. On 1 Hz flights, where the phases are labelled, we take the phase at the instant a flight enters (31 381) or leaves (31 202) a group; entries at the departure (1 179) and exits at the landing (1 358) are counted apart. Both follow the shares of the time alone in each

pilots	hours	median (min)	join	leave	landing	split	merge
2	17 389	41	0.31 [0.30, 0.32]	0.96 [0.95, 0.97]	0.098 [0.094, 0.10]	–	0.085 [0.079, 0.092]
3	3 349	20	0.54 [0.52, 0.57]	1.6 [1.6, 1.6]	0.050 [0.044, 0.056]	–	0.16 [0.15, 0.18]
4	1 059	13	0.70 [0.66, 0.76]	1.6 [1.6, 1.7]	0.042 [0.032, 0.051]	0.48 [0.45, 0.52]	0.26 [0.22, 0.29]
5	421	11	0.88 [0.80, 0.96]	1.7 [1.6, 1.8]	0.036 [0.021, 0.057]	0.82 [0.76, 0.88]	0.32 [0.26, 0.37]
6	176	8	1.1 [0.93, 1.2]	2.1 [1.9, 2.3]	0.028 [0.011, 0.048]	1.2 [1.0, 1.3]	0.51 [0.40, 0.63]
7	96	7	1.1 [0.91, 1.3]	2.0 [1.8, 2.2]	0.031 [0.0089, 0.074]	1.3 [1.1, 1.5]	0.40 [0.30, 0.53]
8+	107	4	1.8 [1.5, 2.2]	2.5 [2.2, 2.8]	0.028 [0.0084, 0.057]	2.7 [2.4, 3.1]	1.0 [0.73, 1.2]

Table 8: By size: hours spent, median duration of a group, and groups ending per hour by outcome, with 5–95% bands.

phase (Table 16 in Appendix H): there is no phase where groups preferentially form or break. 34% of the entries are within 10 min of the departure.

When in the day. Group time is concentrated around midday: 52% of a group flight's time at 12 h, 38% at 9 h, 22% at 18 h (Figure 46 in Appendix H).

7.2 What keeps two pilots together

Do two pilots in a group keep the same pace, or are they pulled back to each other? We follow the delay $\ell^*(t)$ inside each contact (Appendix B), leaving out the instants with either pilot in a search. Groups are the 34 082 pair intervals of pilots who took off within 10 km; chance is the 134 202 contacts of the surrogate pairs, two pilots who never saw each other, and the 980 of these that last at least τ^*, selected as the groups are. Three measures:

- $D(s) = \langle(\ell(t+s) - \ell(t))^2\rangle$, the mean square change of the delay after s, fitted by an Ornstein–Uhlenbeck process, $D(s) = 2\sigma^2(1 - e^{-ks}) + c$: $1/k$ is how long the delay remembers its value, σ its spread, $2k\sigma^2$ how fast it wanders;
- the drift $\langle\ell(t + 60\text{ s}) - \ell(t)\rangle/60$ s at each ℓ, made symmetric since the order of a pair is arbitrary: the pull back to $\ell = 0$, a spring $-k\ell$ near 0;
- the distance between the two pilots at the same instant after their first contact. Two independent pilots separate as twice the displacement variance of one, since the drift along the route cancels in their difference; the surrogate pairs give this baseline on the same terrain.

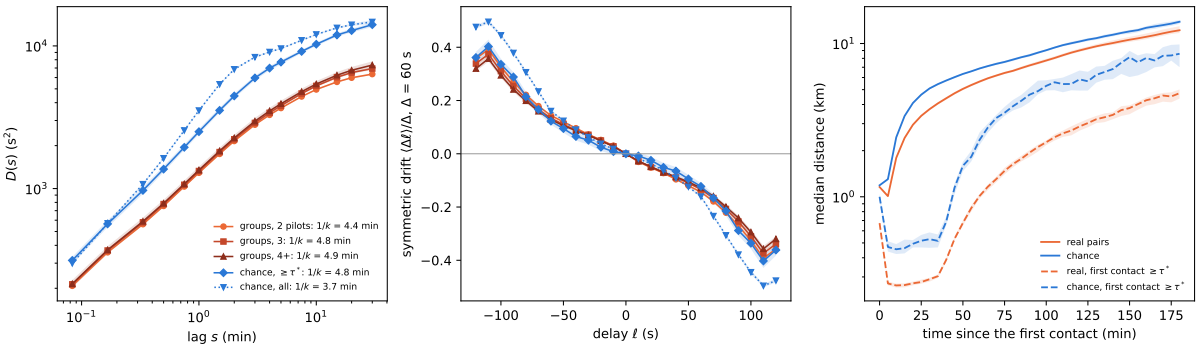


Figure 22: Left: $D(s)$ of groups (by size) and of chance contacts, with $1/k$ of the Ornstein–Uhlenbeck fit. Middle: symmetric drift of the delay. Right: median distance at the same instant after the first contact, for every pair and for those whose first contact lasts at least τ^*. Bands: 5–95% of 200 draws of the dates.

- **Groups keep the same pace.** Their delay wanders at $2k\sigma^2 = 21.4$ [20.9, 22.0] s^2/s, against 43 [41, 46] for chance contacts as long and 62 [61, 63] for all of them; its spread is $\sigma = 55$ s [54, 55], against 79 s [78, 80]. Size does not matter: $2k\sigma^2 = 21.2$, 21.4 and 21.8 s^2/s at 2, 3 and 4+ pilots.
- **They are not pulled back much harder.** The delay forgets itself in $1/k = 4.7$ min [4.6, 4.7], against 4.8 min [4.5, 5.1] by chance; the drift gives 6.9 min [6.7, 7.0] against 7.5 min [6.9, 8.1]. Most of the pull is the selection: a contact that lasts keeps $|\ell| \leq \delta$ by definition.
- **They stay together after the contact that selected them.** When the first contact lasts at least τ^*, the two pilots are 1.0 km apart 60 min later (median), against 2.4 km by chance; 2.9 against 5.8 km after 2 h. Over every first contact, 3.7 against 5.1 km after 30 min.

What holds a group together is a shared pace; a spring adds little to it. Whether the shared pace is following or the same choices in the same air, the data cannot tell.

7.3 Who leads

Who is ahead in a group, how often does that change, and who is it? The answer starts from the shape of the group.

Shape. Inside a group the members are fixed, but the pair intervals among them start and end: each edge set is a configuration. A configuration has a density, a diameter and a shape: complete, a tree (a chain or a star), or between (Appendix F). The TA-MSD pieces of the report *Group and solo flights* lie inside one group and take the shape of its configuration at their middle.

- **Large groups are strings along the route.** Three pilots form a triangle 53% of the time and a chain 47%. Complete graphs fall from 27% of the time at 4 pilots to 2.5% at 8+, the mean density from 0.84 at 3 to 0.45 at 8+; at 8+ the diameter is 4 or more 61% of the time: pilots follow each other along the route.
- **The shape does not change the flight.** $H_{\mathrm{loc}}(1000\ \mathrm{s})$ is 0.906 [0.901, 0.910] in triangles and 0.907 [0.902, 0.911] in chains; 0.915 [0.905, 0.923] in complete graphs of 4+ and 0.907 [0.894, 0.920] in trees. The straighter progress found in *Group and solo flights* does not need every pilot in contact with every other.

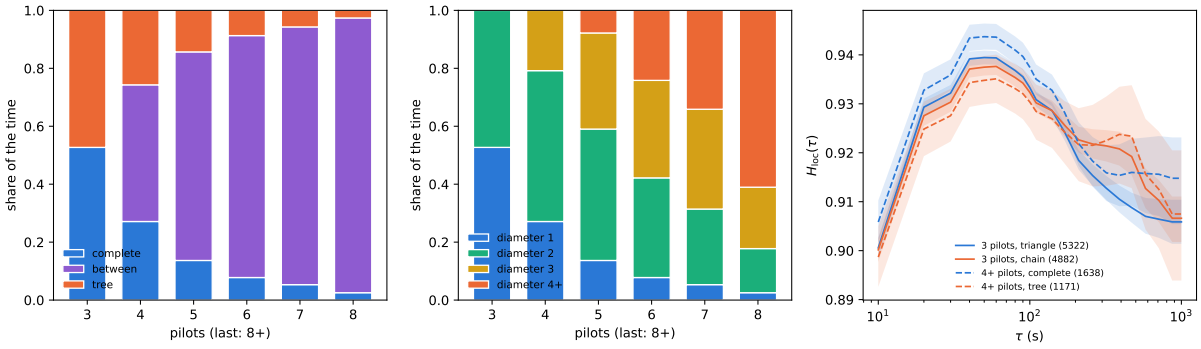


Figure 23: Left: share of the time of each shape, by size (8+ pooled). Middle: share of the time by diameter. Right: $H_{\mathrm{loc}}(\tau)$ of paragliders in triangles and chains of 3 pilots, and in complete and tree graphs of 4 or more (flights in brackets), with 5–95% bands.

Order. The signed median delay ℓ_{med} of a pair interval (Appendix B) says who passes first: the pilot ahead, when the other passes at least 15 s later; otherwise the two fly abreast (33% of the pairs of two; median delay 20 s). How leading changes the flight is in *Group and solo flights*; here we ask who leads (Figure 24; Figure 47 in Appendix H).

- **Pilots are in a line.** In groups of 3 pilots or more, each edge points from the pilot ahead to the one behind; we compare with the same graphs with each direction drawn at random (50 draws). Directed triangles are cyclic 2.3% [1.8, 2.8] of the time, against 25% at random; graphs with every edge directed (34% of the time) are acyclic 98.7%, against 88%. Positions along one route cannot form cycles, so the order is geometry and says nothing about rank.
- **One leader, in small groups:** 80.9% [80.0, 82.0] of the acyclic time, against 77.2% [76.7, 77.8] at random; 86% at 3 pilots, 68% at 4, 52% at 5 or more. A single file (depth $n-1$): 71, 47 and 26%.
- **The lead changes often.** Along $\ell^*(t)$, with stretches shorter than 60 s dropped, the pilot ahead changes 3.3 times per hour of contact [3.3, 3.4], against 4.3 [4.1, 4.4] in chance contacts lasting at least τ^*; the pilot most often ahead holds 74% of the directed time (median), against 69%.
- **The experienced pilot on the faster wing leads.** In groups of two named pilots, the leader has more earlier flights in the catalog in 58.5% [57.8, 59.1] of 16 294 pairs (1.3 times as many, median), and the higher wing class (A = 1 to competition = 5) in 64.3% [63.4, 65.3] of 8 475.
- **Some pilots lead or follow persistently.** For the 975 pilots with at least 10 roles in pairs, the dispersion of their lead fractions is 2.73 against 1.00 with roles swapped at random (95% below 1.08); 12% lead in at most 20% or at least 80% of their roles, against 3% expected.

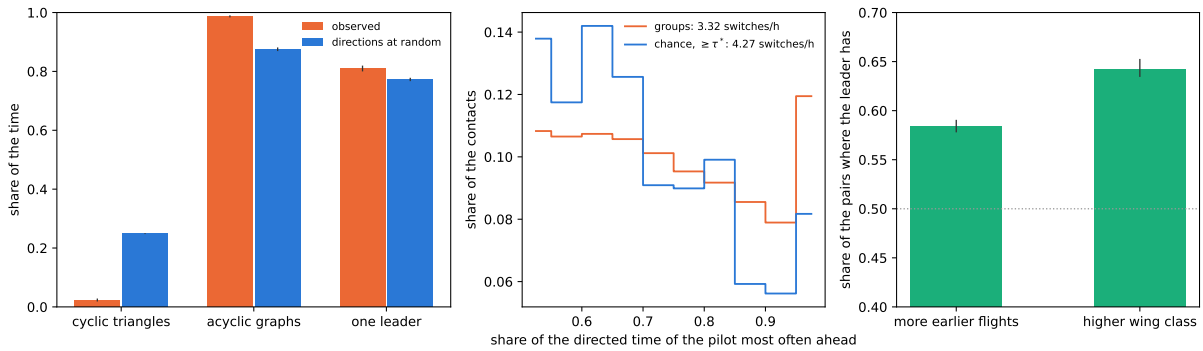


Figure 24: Left: in groups of 3 or more, share of the time with cyclic directed triangles, acyclic graphs (every edge directed) and one leader (acyclic graphs), observed and with directions at random. Middle: share of the directed time of the pilot most often ahead in each contact, with the switches per hour. Right: share of the pairs of two in which the leader has more earlier flights, or the higher wing class. Bars: 5–95% (blocks).

7.4 With whom

Do pilots choose their partners, or fly with whoever is there? We follow paraglider pilots by their name in the catalog (anonymised ones left out). Two pilots are partners on a day when

their flights share a pair interval: a *pair-day*. Chance keeps who flies in a group where and when: within each site-day (takeoffs within 3 km, joined when a pair interval links two sites), the partners are redrawn at random, each pilot keeping its number of partners that day (a configuration model; 100 draws). There are 30 550 pair-days among 4 525 pilots and 22 513 pairs; one pair flew together on 69 days (Figure 25).

- **Pilots return to the same partners:** 37.6% of the pair-days are on pairs that meet on more than one day, against 25.7% [25.3, 26.0] at random (3 437 such pairs against 2 501).
- **and to their club:** 32.1% of the pair-days join two pilots of the same club, against 24.0% [23.8, 24.2].

Chance already fixes who flies in a group at each site and day, so the excess is the choice of partner among the pilots there. Choosing where and when to fly with friends stays in chance here.

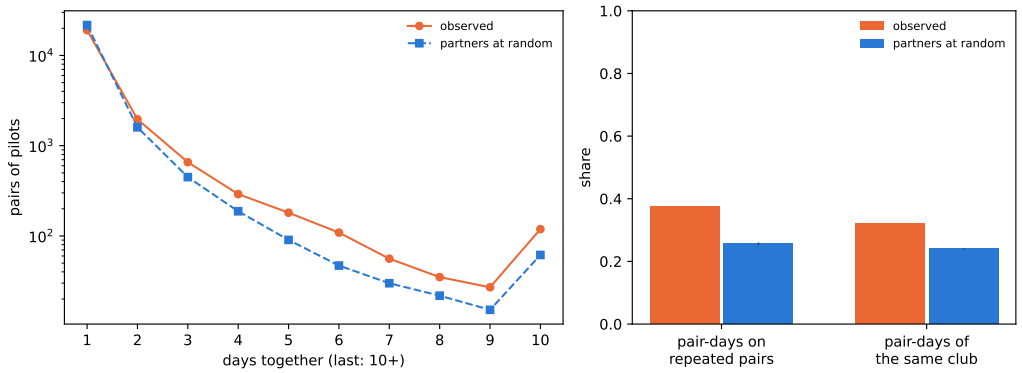


Figure 25: Left: pairs of pilots by the number of days they flew together, observed and with partners at random. Right: share of the pair-days on pairs that meet on more than one day, and of two pilots of the same club (each pilot's main club); bars at random: 5–95% of the draws.

7.5 Where the sizes come from

Why are most groups pairs, and how do a few grow large? $T(n)$ is the share of the time spent in groups of n pilots: the chance that a group seen at a random instant has n pilots. If sizes change one pilot at a time, the flux from n to $n+1$ pilots balances the flux back, a birth–death chain (Appendix G):

$$\frac{T(n+1)}{T(n)} = \frac{g(n)}{l(n+1)}, \tag{13}$$

with $g(n)$ the joins and $l(n)$ the leaves and landings per hour at size n (Table 8); splits and merges are left out. Per member, $g(n) = a\,n$ and $l(n) = \mu\,n^{1-\gamma}$: every member brings in others at rate a and leaves at rate $\mu n^{-\gamma}$. We fit these to the rates, without looking at T. With $\gamma = 0$ the chain gives the log-series $T(n) \propto x^n/n$, $x = a/\mu$. For comparison, we fit the form of Bonabeau et al. [6], $T(n) \propto n^{-\alpha} e^{-n/n_c}$, directly to T.

- **One pilot at a time is enough.** With the measured rates the chain gives $T(n)$ within 4% from 2 to 8 pilots (0.145 against 0.148 at 3 pilots, just outside its band), although splits and merges are 6% of the changes at 2 pilots and 46% at 8+.
- **Members join at a constant rate and leave less often in large groups.** Each member brings in others at $a = 0.162$ [0.158, 0.167] per hour, at every size. Each leaves at 0.53 per hour in a pair and 0.25 at 8+: $\gamma = 0.30$ [0.28, 0.33].

- **The heavy tail comes from staying.** With $\gamma = 0$ ($x = 0.25$) $T(10)$ is 290 times too small; with $\gamma = 0.30$, 4 times. Fitted directly, $\alpha = 3.4$ and $n_c = 5$.

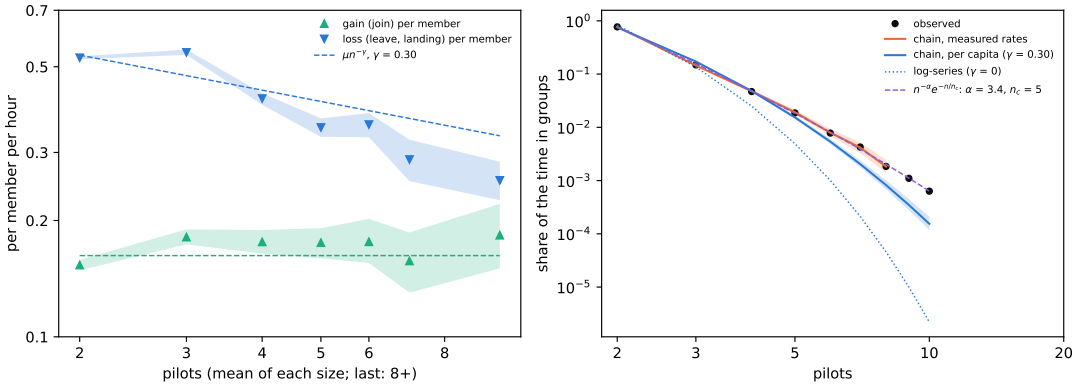


Figure 26: Left: joins and losses per member per hour by size (mean size of each class; 8+ pooled), with a and $\mu n^{-\gamma}$ dashed. Right: $T(n)$ observed (sizes with at least 10 h) and from the chain (13) with the measured rates, with the per-capita rates and with $\gamma = 0$; dashed purple: the form of Bonabeau et al. fitted to T. Bands: 5–95%.

7.6 What the data support

Supported

- groups are mostly pairs; large ones are strings along the route that change members every few minutes, while the episode lasts about an hour;
- group pilots keep the same pace: their delay wanders half as fast as that of two pilots kept in contact by chance, and they stay closer after the contact;
- the size distribution follows from one pilot joining or leaving at a time; its heavy tail from members of larger groups leaving less often;
- the lead changes a few times per hour and goes more often to the experienced pilot on the higher wing;
- pilots return to the same partners and to their club mates more than who flies where and when explains.

Not supported

- a spring that pulls pilots back: the pull is about that of chance contacts selected the same way;
- that the shape of a group (triangle or chain, complete or tree) changes how its pilots fly;
- a social rank from the order in a group: a line along the route is geometry;
- that keeping pace means following: the surrogate pairs fly at another hour, so pilots in the same air at the same time may share the pace without seeing each other;
- pilots are names of the catalog: homonyms join, a pilot who changes name splits.

One caveat: in a thermal the altitude fixes $\ell^*(t)$ to about a turn, 20 s (Appendix B), for real and chance contacts alike, so the cohesion and the switches of the lead carry this noise on both sides of the comparison with chance.

8 Use and limits

What are the labels for, and what must the thesis declare when it uses them? They exist to compare how pilots move in a group and alone; this section says how, which choices come with them, and what has been checked.

8.1 Comparing group and solo flight

Labels on the phases. For a run σ of one phase (climb, search or glide) of flight i, with interval J_σ, we define the fraction in a group and the mean size:

$$f_\sigma = \frac{\left|\{t \in J_\sigma \cap T_i^{\text{dep}} : \gamma_i(t) \notin \{\mathsf{L}, \mathsf{S}\}\}\right|}{|J_\sigma \cap T_i^{\text{dep}}|}, \qquad \bar{n}_\sigma = \text{mean of } n_i(t) \text{ over } J_\sigma \cap T_i^{\text{dep}}, \tag{14}$$

with $n_i(t)$ the number of members of the group of i at t, 1 if solo. The run is *in a group* if $f_\sigma \geq 1/2$ and *solo* if $f_\sigma = 0$; the intermediate cases are excluded, or f_σ is used as a continuous covariate (Figure 27). Runs in the local phase are excluded.

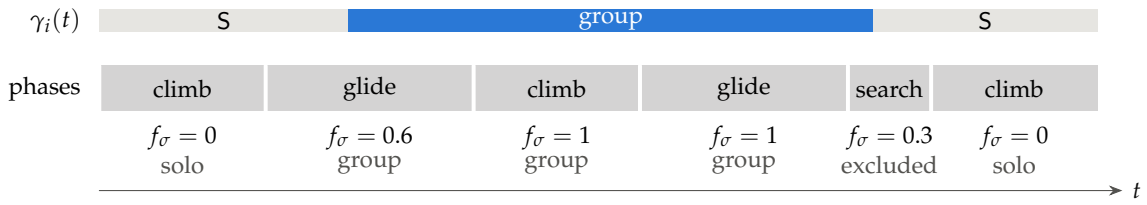


Figure 27: Labels on the phases of one flight: each run of one phase gets the fraction f_σ of its duration spent in a group, and from it a class.

Stratification. Group flights concentrate on the good days, at the busy sites and around midday (Section 7.1), so a global comparison would mix the effect of the group with that of the weather. We compare within strata s (date and area):

$$\hat{\Delta} = \sum_s w_s \left(\hat{\theta}_{G,s} - \hat{\theta}_{S,s}\right), \tag{15}$$

where $\hat{\theta}$ is the statistic of interest and the weights w_s are proportional, for example, to the number of group runs of the stratum.

Uncertainties. Runs of the same flight are not independent, and neither are those of pilots of the same episode. The bootstrap resamples the dependency blocks (Section 4.5) within the strata.

Confounders. Competition days; the level of the pilot (the same pilots often fly together, Section 7.4); the glider class (EN A–D); the time of day.

Questions to ask. Distributions of the times spent in thermals and of the lengths of the glides, in a group and alone; mean squared displacement and effective exponent; differences between followers and leaders, by the sign of ℓ_{med}. In birds, followers leave the thermals earlier and lower [9], and individuals with social information glide faster [31]: these are hypotheses to test on pilots before assuming them.

8.2 Choices to declare and limits

The starting values and those chosen on the data are in Table 2. The thesis should state these choices, each argued in the section given:

- ε_z encodes the notion of "together": shared thermals if tight, visual contact if wide (Section 3.4);
- δ is bounded by the visual range through v_{xc} (Section 3.4) and measured on the data at 2 min; $q = 0.1$ lets at most one contact in ten be chance; ε_h, ε_z and g show no plateau and are definitions (Section 6);
- headings are compared only when both pilots glide; climb, search and unclassified fixes turn the condition off (Section 3.3); in the run reported here the phases are Vilpellet's, on 1 Hz flights only, so the condition acts only between two such flights (Section 2);
- contacts before the departure are ignored: a group formed on the takeoff is recognised only after the exit from R_{dep} (Section 3.1);
- grid positions are interpolated across at most $h_{\max}$, so tracks logged more coarsely enter with gaps (Appendix C.2);
- the signed median delay leaves out the instants of search, where the delay is not determined (Appendix B);
- chance is defined by the null model, whose shifts of about an hour also remove the synchrony of pilots with the weather of the hour: what it calls real includes pilots in the same thermals at the same time who do not follow each other, the "same air" notion of Remark 3.3, which is no proof of interaction; τ^* is 23 min with shifts over an hour and 53 min with shifts under 30 min (Section 5.3);
- the errors of the phase labelling reach the heading condition: a climb labelled as glide compares headings that turn, and joining the short gaps mitigates the problem.

The measurements have three further limits:

- the profiles of Sections 6.1 and 6.2 are on one sample of 300 dates;
- the plateau check of Section 6.3 is on 300 dates and with the starting delay of 5 min;
- the data are the cleaned tracks of the old repository (soaring, tag `legacy-final`), not the pre-processing of xc-dynamics that Section 2 describes.

8.3 Checks done

- **Code against this note.** Every phase was compared with its code (grid, departure, candidate filter, scan of delays, contact, gaps, signed delay, null model, groups, episodes, membership, blocks): all implemented as written, with four differences of text, now corrected in red: (1) the phases of the run are Vilpellet's, on 1 Hz flights, so the heading condition acts only between two such flights; (2) headings are in the east and north of the first fix of each flight, not of the fix itself; (3) a contact is kept only from the later departure to the earlier landing; (4) instants in a gap of the track are labelled solo in the tables.

- **Properties of the tables** of the chosen run: every pair interval lasts at least $\tau^* = 2285$ s and lies where both flights fly after departure; every group has at least two members; membership covers each flight from takeoff to landing without holes; on 3 000 flights the group time equals the union of its pair intervals; on 3 000 groups the members are connected at the midpoint and no active interval leaves; on 2 000 episodes every group but the first starts when another ends and shares a member; the configurations of every group keep its members and follow one another over its interval without gaps; every episode lies in one block. The counts of Table 6 (154 598 and 6 039 flights; 21.7% and 6.9% in a group) are reproduced.

8.4 Checks to do

- Visual inspection: about ten episodes on a map, with the entries and exits checked by hand.
- Sensitivity to the step: on a subset of pairs, the computation with $\Delta = 1$ s must give the same intervals.
- Competition days: if the dataset contains competition tracks, on those days everybody flies the same course and is in a group by construction; they are flagged and treated apart.
- Grid: the share of grid instants without a position, per flight and by logging interval.

What we keep. The labels enter the comparison phase by phase, within strata of date and area, with the dependency blocks as the units of the bootstrap, and with the choices of Section 8.2 declared, first among them what chance means.

A Relation to the literature

Where does each ingredient of the method come from? This appendix reviews the precedents, as far as a targeted but not exhaustive search shows. The introduction summarises what we take and what is new (Section 1), and Table 9 details it.

A.1 Trajectory similarity with a temporal tolerance

The LCSS of Vlachos, Kollios and Gunopulos [30] matches two points when they are closer than ε in every coordinate and their *indices* in the two sequences differ by at most δ. The similarity S_1 is the length of the common subsequence divided by the length of the shorter track, and the variant S_2 maximises S_1 over all spatial translations, to recognise parallel movements. The EDR of Chen, Özsu and Oria [8] and DTW with a Sakoe–Chiba band [23] bound the shift in the same way. These measures return a single number per pair of tracks, so they cannot say *when* two pilots are together. A bound on indices counted from the first point of each track also mixes the shift between the takeoffs with the delay in flight, and S_2 matches precisely the parallel routes that we want to exclude.

The measures of joint movement that do say when are built on simultaneous fixes. The proximity index Prox is the share of simultaneous fixes closer than a threshold [13, §3.1]: with $\delta = 0$ and without the conditions on height and heading, it is the share of the flight of i in contact with j in the sense of Section 3.2.

A.2 Following with a delay

The closest precedent of our pair contact is the following of Li, Wu and Crofoot [16], defined on tracks sampled at the same instants (Figure 28). At instant i the local lag $\Delta(i)$ is the shift t, with $|t| \leq l_{\max}$, that minimises the distance between the position r_i of R and the position s_{i-t} of S. The instant is *valid* if this minimum distance is below $d_{\max}$, and R follows S at a valid instant if $\Delta(i) > 0$ (§IV-A). The lag is therefore local and free to vary, and its sign separates the follower from the leader. Two individuals that merely move together lead and follow equally often, so the following intervals are the stretches with more following instants than the half expected without following: the maximal segments of $f(i) - 1/2$, with $f(i) = 1$ when R follows and 0 otherwise, found in linear time [22] (§IV-B) and checked with the roles reversed (§IV-C). The threshold $d_{\max}$ only discards the instants without interaction, and does not move the minimiser.

Our contact keeps the window of lags and the distance threshold, and leaves the sign out: the contact of i with j at instant t is the validity of t, with tolerances on height and heading added (Section 3.2). We want the pairs that fly together, whether or not one of them leads; who follows whom comes back afterwards, through the median lag of each contact (Appendix B).

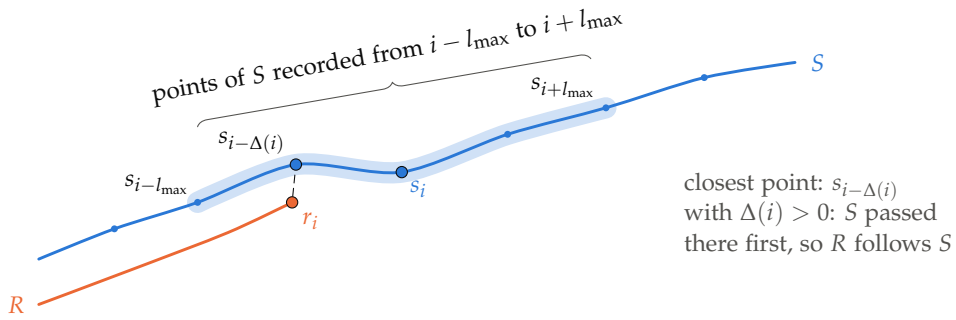


Figure 28: Following with a local lag [16]. At instant i, the track of S is searched in the time window $[i - l_{\max}, i + l_{\max}]$ (shaded) for the point closest to the position r_i of R. The best lag $\Delta(i)$ is positive: R is now where S was $\Delta(i)$ earlier.

Nagy et al. [19] measure the delay between two pigeons as the shift that maximises the correlation of their headings, and build leader–follower hierarchies from its sign. In a thermal the heading turns through 2π every 20–30 s, so this correlation, like the distance of Section 3.3, repeats with the period of the turn. Konzack et al. [14] display such local delays between simultaneously recorded tracks in a "delay space", comparing Fréchet, DTW and edit-distance matchings. FLICA and its package mFLICA [2, 3, 4] infer a following network in sliding time windows, with DTW restricted to a Sakoe–Chiba band, and call the faction of a leader the individuals that reach it through a directed path: groups built from delayed following, but directed, with leaders, and on windows. The following of Andersson et al. [5] is instantaneous: an individual follows another that is in a sector ahead of it with a similar heading.

A.3 Group structure in time

Buchin et al. [7] call two entities directly connected at time t if their ε-discs overlap, and ε-connected if a chain of direct connections joins them at that instant; the maximal ε-connected sets, the components, partition the entities at every instant. The Reeb graph records how the components evolve: its vertices are the events (start, end, merge, split), and each edge is a fixed set of entities that forms one component over an interval of time (Figure 29). On it they define the maximal groups, sets of at least m entities that stay inside one component for at least a duration δ; maximal groups may overlap, and an entity can belong to several at once. Two

more ideas of theirs reappear in the method: a robust structure ignores interruptions shorter than α, by calling two entities connected at t when they were directly connected at some instant within $\alpha/2$ of t (§4); and the grouping structure can split the trajectories into independent data sets (§1). The proximity is always simultaneous.

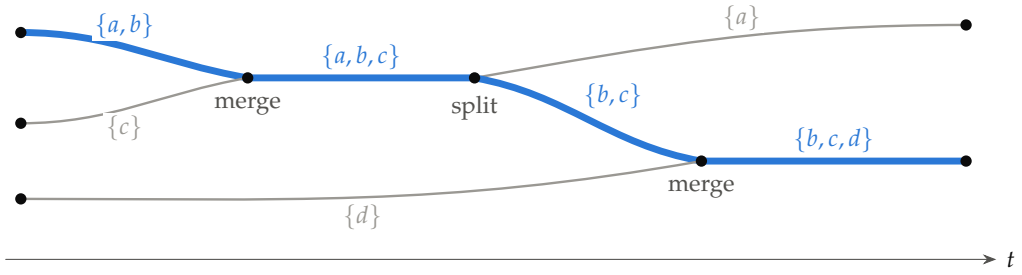


Figure 29: The grouping structure of Buchin et al. [7] as a Reeb graph. Each edge is a fixed set of entities that forms one connected component over an interval; vertices are starts, ends, merges and splits. The thick edges have at least two entities: in this note they become the groups (Section 4.3), built on delayed contacts.

A.4 Interaction indices and null models

Joo et al. [13] sort the metrics of joint movement between pairs by what they capture, proximity or coordination. Long et al. [17] compare eight of them on simulated tracks and on deer: the indices that rest on a statistical test report interactions that are not there, more often the finer the sampling, and the indices that return one number per pair miss the interactions that come and go; the wildlifeDI package [18] extends the analysis to contacts in high-frequency data. Spiegel et al. [24] build a null model by shuffling whole days of each individual's path among its days: each individual keeps its use of space, and the synchrony between individuals is lost.

A.5 Collective soaring

Ákos, Nagy and Vicsek [1] compare the soaring of birds with that of paraglider and hang-glider pilots on GPS tracks; falcons, storks and paragliders circle with radii of 20–26 m. In migrating white storks a few leaders find and explore the thermals and the others follow them, leaving the thermals earlier and lower [9]; inside a thermal, flock mates synchronise their circling [20]. Vultures informed by other birds of the next updraft glide towards it faster [31]. We found no work that identifies group flights systematically in large free-flight datasets.

A.6 What we take from each precedent

Table 9 lists, component by component, the precedent, what we take from it and what we change; the three points we did not find in the literature are in Section 1.

Essential reading. The two readings to do first are Li et al. [16], Sections II and IV (following, local lag, score), and Buchin et al. [7], Sections 1, 2.1 and 4 of the arXiv version (groups, Reeb graph, robustness). Long et al. [17] and Spiegel et al. [24] serve the null model; Vlachos et al. [30] and Joo et al. [13] place the measures of similarity and proximity.

Component	Precedent	What we take	What we change
Pair contact with a delay	Li et al. [16]; tolerance in [30, 8, 23]	window of lags and distance threshold; local lag, free to vary	height tolerance (separates heights within a thermal); heading only in glide; local phase excluded; no sign in the contact, $a_{ij} = a_{ji}$ (the sign returns in ℓ_{med})
Contact intervals	robustness in [7]; maximal segments in [16]	tolerance to short interruptions	gaps shorter than g filled; minimum duration calibrated on a null model
Groups in time	Buchin et al. [7]	components and their Reeb graph; groups are its edges with at least two entities	built on delayed contacts; episodes are the connected parts of the Reeb graph without single entities; no maximal groups
Groups from following	mFLICA [4]	–	undirected groups without leaders; exact intervals instead of sliding windows
Null model	Spiegel et al. [24]; SAM [28]	time shift of the tracks; false discovery rate as chance count over observed count	shift within the day, drawn from the real delays between takeoffs; chance count per unit of exposure
Computation	filter–refine scheme	–	no-loss lemma for delayed contacts; KD-tree with a time coordinate

Table 9: Where the ingredients of the method come from.

B The delay of a contact: who follows whom

The groups of Section 4 are built from the contact alone. To say who follows whom (Sections 7.3 and 8.1) we also need how many seconds j is behind i. We estimate it twice: at each instant of a contact, and over the whole contact.

The delay at one instant. At an instant t with $m_{ij}(t) = 1$, pilot i has at least one correspondence $(t, t + l) \in \mathcal{M}_{ij}$: a delay l at which j is where i is, within the tolerances. Of these we take the delay at which j is closest to i in space, with the altitude added to the closest point of [16],

$$l^*(t) = \arg \min_{l:\,(t,t+l)\in\mathcal{M}_{ij}} \sqrt{\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|^2 + \big(z_i(t) - z_j(t+l)\big)^2}, \tag{16}$$

so $l^*(t) > 0$ when j passes later where i was. How well it is defined depends on the phase of flight.

The delay of a contact. Over a pair interval $[t_{\text{on}}, t_{\text{off}})$ (Definition 4.1) the delay is the signed median of $l^*(t)$,

$$\ell_{\text{med}} = \text{median}\{l^*(t) :\ t \in [t_{\text{on}}, t_{\text{off}}),\ m_{ij}(t) = 1,\ \text{neither pilot in a search}\}, \tag{17}$$

with $\ell_{\text{med}} > 0$ when j follows i. It says who follows, and it does not enter the groups. It is undefined when every instant is in a search.

Glide. The path is nearly straight, and a second of delay moves j by about 10 m along it and changes its altitude by about 1 m: the distance has a single sharp minimum, and l^* is the delay at which j crosses the point of i (Figure 30, left).

Climb. j circles over one spot, so the horizontal distance repeats at every turn and has an equal minimum at each, about every 20 s: it cannot tell which turn is the right one. The altitude can, since j climbs and passes the altitude of i once. In the 3D distance the position on the circle sets the dips and the altitude sets their depth, so the deepest dip is the turn at which j is at the height of i and at its point of the circle (Figure 30, right). The altitude changes by tens of metres from one turn to the next, as much as the radius of the circle, so at one instant l^* is known to about a turn.

Search. The pilot turns without climbing: the altitude does not change with the delay, and the 3D distance has, as the horizontal one, an equal minimum at every turn and no way to choose among them. The instants at which i at t or j at $t + l^*$ is in a search are left out of the median.

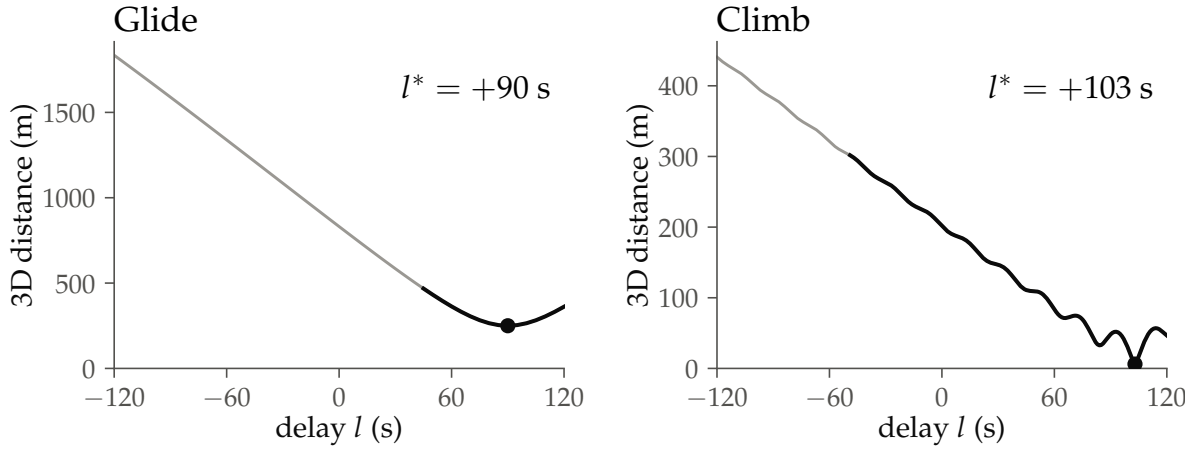


Figure 30: The delay at one instant t, for every delay l of $[-\delta, \delta]$, $\delta = 2$ min. The curve is the 3D distance between i at t and j at $t + l$; the black part marks the delays that are correspondences ($\varepsilon_h = 500$ m, $\varepsilon_z = 300$ m) and the dot the minimum, l^*. Left: j flies the route of i 90 s later and 250 m to the side, at the same altitude: one minimum, at $l^* = +90$ s. Right: i and j circle in one thermal, 25 m in radius, one turn every 20 s, both climbing at 2 m/s, j lower by 200 m. The distance dips at every turn; the deepest dip, at $l^* = +103$ s, is where j has climbed the 200 m (100 s) and reached the point of the circle where i was (3 s more).

What we keep. At each instant of a contact, the delay $l^*(t)$ is the minimum of the 3D distance between i at t and j at $t + l$ over the correspondences. Over a pair interval, ℓ_{med} is the median of $l^*(t)$ outside the searches. In a glide l^* is the delay at which j crosses the point of i; in a climb the altitude chooses the turn; in a search nothing chooses, and the instant is left out.

C Algorithm

How can the definitions of Sections 3 and 4 be computed on the whole archive? Literally, for every pair of flights of a day, for every instant of one and every delay up to δ, one would test

the four conditions of (2). On a day with 200 flights in the air this is $200 \cdot 199/2 = 19\,900$ pairs; with 1.5 h of flight on the 5-s grid, 1 080 instants, and the 49 delays of $\delta = 2$ min, it is about 10^9 tests for that day alone. Almost all of them are wasted: most pairs of the day are never within several kilometres of each other. The algorithm therefore works in phases (Figure 31): a common time grid (Phase 0); a cheap test that discards, provably losing nothing, the pairs and instants without a correspondence (Phase 1); the exact test on what remains (Phase 2); the intervals (Phase 3); and a sweep over time that builds the groups (Phase 4).

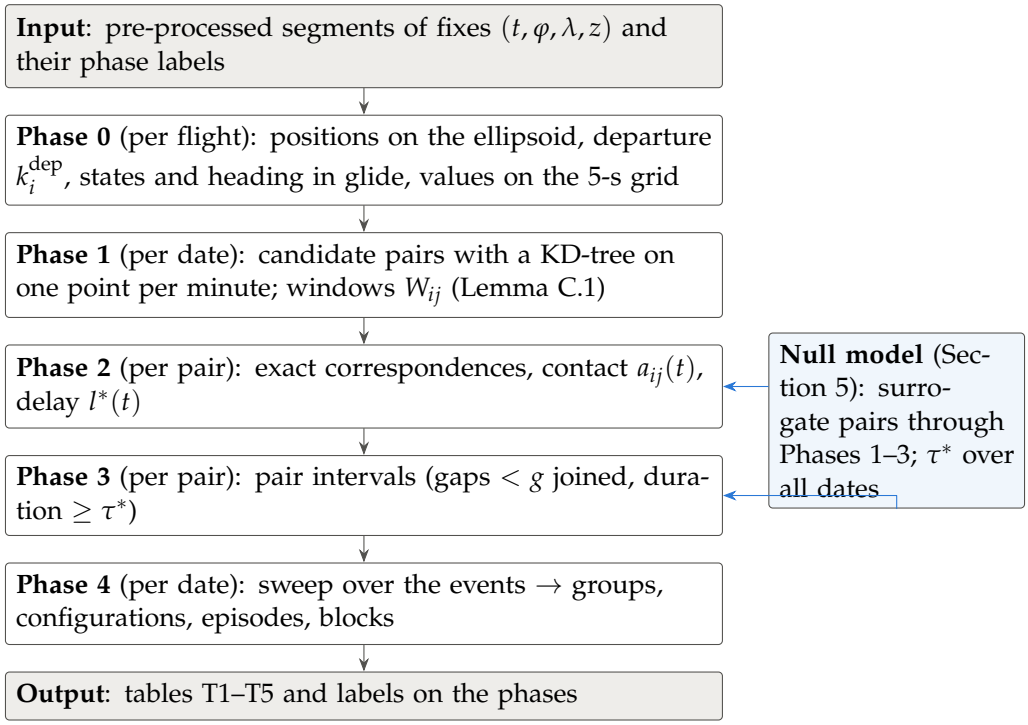


Figure 31: The method as a pipeline. Phases 0–3 and the null model run date by date; τ^* is computed once, over all dates; Phase 4 then runs date by date with the intervals lasting at least τ^*.

C.1 How a run is organised

Flights of different days never meet, so every phase but the calibration works on one date at a time: the dates run in parallel, and the memory needed is that of one day. A run has three stages.

1. **Per date.** Phase 0 puts the flights of the date on the grid; Phases 1–3 find the contact intervals of the real pairs, with their duration and ℓ_{med}, but without the threshold τ^*, which is not known yet. The null model of Section 5.1 draws the surrogate pairs of the date and sends each one through the same Phases 1–3, with the copy of j shifted in time and filtered against i alone. The result of each date is saved, so an interrupted run resumes.

2. **Once.** τ^* follows from the real and the surrogate intervals of all the dates, Eq. (10).

3. **Per date.** Phase 4 keeps the intervals lasting at least τ^* and builds groups, configurations, episodes and blocks.

On the whole archive, 6 209 dates, a run takes about an hour with five processes.

C.2 Phase 0: preparation per flight

(a) **Positions.** The Earth-centred position of every fix (Section 2). One frame serves all the flights, at any distance and without a map projection, so the coordinates of two flights compare directly.

(b) **Altitude.** GNSS z, unchanged.

(c) **Departure.** k_i^{dep} from (1).

(d) **States and heading.** $s_i(k)$ from the phase labelling. In glide, $\hat{\mathbf{v}}_i(k)$ is the direction of the horizontal velocity, filtered with Savitzky–Golay, in the east and north of the first fix of the flight (the frame of the cleaned tracks) instead of those of the fix itself. So two flights are compared in frames whose axes differ by the rotation of the local north between their first fixes and the points compared: $(d/R)\tan\varphi$ for a displacement d to the east, about 3° for 300 km at latitude 45°, small against $\theta_{\max} = 45°$.

(e) **Grid.** The instants of $\Delta\mathbb{Z}$, with $\Delta = 5$ s, the same for all flights. A grid instant takes the position and the altitude of the fix at that instant or, between two fixes of the same segment at most $h_{\max} = 10$ s apart, the values interpolated linearly between them, with the state and the heading of the nearer fix; elsewhere the flight has no position. On a logger that records every second every grid instant is a fix. An interpolated position is at most $v_{\max}h/2$ from the true one, for two fixes h apart: 69 m at 5 s, 139 m at 10 s.

Is $\Delta = 5$ s enough in a thermal?

Four points per turn do not draw the spiral (Figure 32), but the contact does not need its shape. What matters is how far a grid position can be from the track between two grid instants, compared with the tolerances:

- horizontally, a pilot is never more than $v_{\max}\Delta/2 \approx 69$ m from the nearest grid point in the worst case, about 20 m while circling at 8 m/s; $\varepsilon_h = 500$ m, and a whole thermal of radius 25 m fits in a small fraction of it;
- vertically, $w\Delta/2 = 5$ m at 2 m/s and at most 12.5 m at 5 m/s, against $\varepsilon_z = 300$ m;
- on the delay, a grid of 5 s against $\delta = 120$ s; in a thermal l^* is known to about a turn, 20 s, anyway (Appendix B);
- on the ends of the intervals, $\pm\Delta$ against $g = 120$ s.

States and headings are computed on the fixes, before the grid, and lose nothing. The check of Section 8.4 repeats the computation with $\Delta = 1$ s on a subset of 1-Hz pairs.

`date`	`flight`	`t`	`X, Y, Z`	`z`	`ve, vn`	`state`
partition	int32	int32 (s)	float32	float32	float16	int8

Table 10: Table of the grid points, in Parquet partitioned by date: about 30 bytes per row. `X, Y, Z` are the Earth-centred position, to 0.5 m in float32; `ve, vn` the unit heading in the east and north of the point.

Under the assumption, to be checked, of 1.86×10^5 flights lasting 1.5 h on average, the table has about 2×10^8 rows: 6 GB uncompressed, less in Parquet. It is worth keeping, because the null model and the calibration read it several times.

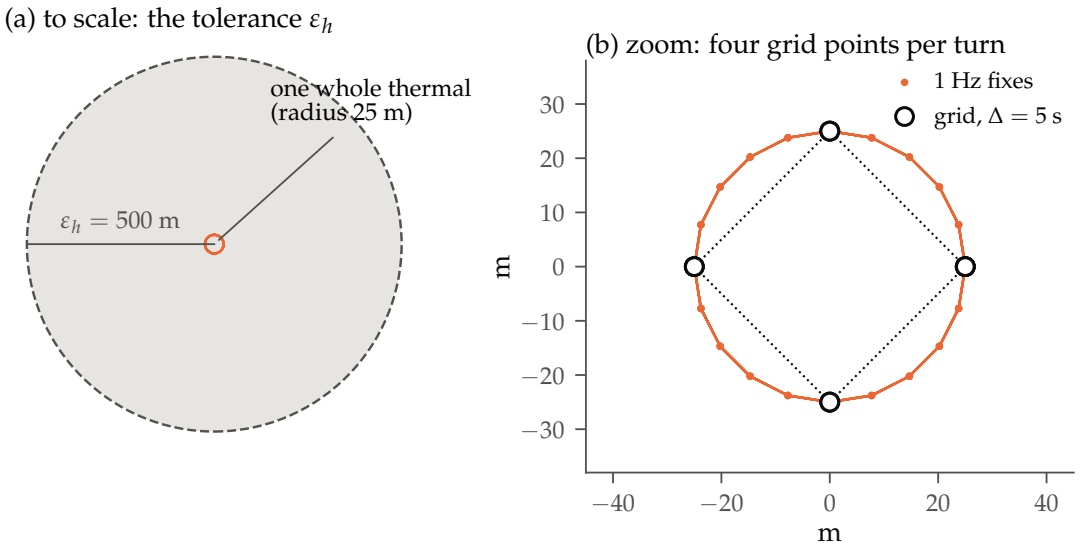


Figure 32: The 5-s grid in a thermal, to scale. (a) The horizontal tolerance ε_h and a whole thermal of radius 25 m. (b) One turn of 20 s sampled at 1 Hz and on the grid of $\Delta = 5$ s: four grid points per turn, each within 20 m of the 1 Hz fixes around it.

C.3 Phase 1: candidate pairs

Phase 1 answers one question for each date: which pairs of flights, and at which instants, can have a correspondence at all? If i at t is within ε_h of j at t', with $|t' - t| \le \delta$, then at the same instant the two pilots cannot be farther apart than ε_h plus the distance j can fly in δ. Testing this at every instant would cost as much as the exact test, so it is tested only at whole minutes, with a radius widened to cover the rounding.

Pieces of track. The bound needs a top speed: between any two instants of a piece of track, the pilot moves at most $v_{\max}$ times the time between them, with $v_{\max} = 100$ km/h, the top speed of the pre-processing. A *piece* is a run of grid positions of one segment of one flight with no step faster than $v_{\max}$; a faster step, which the excursion rule of the pre-processing should already have removed, ends a piece. Let $\tilde{\mathbf{x}}_i(s)$ be a piece of flight i extended by holding its first position before it and its last position after it. The filter reads positions at whole minutes, interpolated linearly between grid instants; a minute may fall just before or just after a piece, and the extension gives the track a position there.

Lemma C.1 (no false negative). *If $(t, t') \in \mathcal{M}_{ij}$, $\tilde{\mathbf{x}}_i$ and $\tilde{\mathbf{x}}_j$ are the extended pieces that contain t and t', and $t_c = 60 \cdot \mathrm{round}(t/60)$, then*

$$\|\tilde{\mathbf{x}}_i(t_c) - \tilde{\mathbf{x}}_j(t_c)\| \le D_c := \varepsilon_h + v_{\max}\,(\delta + 60\ \mathrm{s}). \tag{18}$$

> **In words.** t_c is the whole minute nearest to t (Figure 33a), and both positions in (18) are read at that *same* instant. A pair farther than D_c at every minute has no correspondence and can be discarded.

Proof. The extension with a fixed position keeps the Lipschitz property $\|\tilde{\mathbf{x}}(s) - \tilde{\mathbf{x}}(s')\| \le v_{\max}|s - s'|$ for all s, s'. By construction of t_c, $|t - t_c| \le 30$ s, and $|t' - t_c| \le |t' - t| + |t - t_c| \le \delta + 30$ s. The triangle inequality along the path $\tilde{\mathbf{x}}_i(t_c) \to \mathbf{x}_i(t) \to \mathbf{x}_j(t') \to \tilde{\mathbf{x}}_j(t_c)$ (Figure 33)

gives

$$\begin{aligned}\|\tilde{\mathbf{x}}_i(t_c) - \tilde{\mathbf{x}}_j(t_c)\| &\le \|\tilde{\mathbf{x}}_i(t_c) - \mathbf{x}_i(t)\| + \|\mathbf{x}_i(t) - \mathbf{x}_j(t')\| + \|\mathbf{x}_j(t') - \tilde{\mathbf{x}}_j(t_c)\| \\ &\le 30\, v_{\max} + \varepsilon_h + v_{\max}(\delta + 30),\end{aligned}$$

where the middle term is below ε_h because (t, t') is a correspondence (Definition 3.1). □

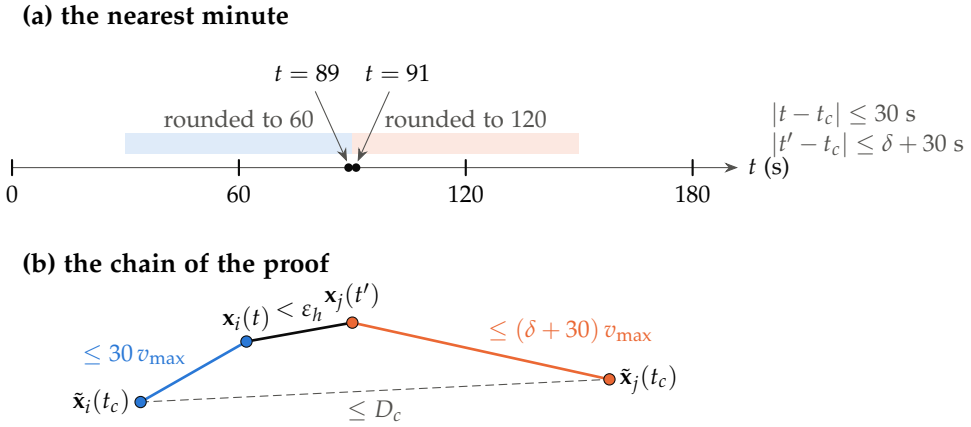


Figure 33: Lemma C.1. (a) Every instant is at most 30 s from its nearest whole minute. (b) From i at the minute to j at the same minute through the correspondence (t, t'): each step is bounded by the time it takes at most $v_{\max}$, or by ε_h.

The radius. With $v_{\max} = 100$ km/h, $\delta = 300$ s and $\varepsilon_h = 500$ m, $D_c \approx 10.5$ km; with the delay chosen on the data, $\delta = 120$ s, $D_c \approx 5.5$ km. The spacing of one minute is a compromise: with points every s seconds the same proof gives $D_c = \varepsilon_h + v_{\max}(\delta + s)$, so a finer spacing narrows the radius a little but multiplies the points to compare.

What a KD-tree is. Phase 1 must find, among the points of one minute, all the pairs closer than D_c. Comparing every point with every other costs $n^2/2$ distances for n points. A KD-tree (a k-dimensional tree) avoids most of them. It splits the points in two at the median of one coordinate, then each half at the median of the next coordinate, and so on, until each cell holds a few points (Figure 34). Every node of the tree is a box of space, and the leaves partition the points. To find the points within D_c of a given point, the search starts at the root and descends only into the boxes that the ball of radius D_c reaches; a box farther than D_c from the point is skipped with all its points, which are never compared. Building the tree takes $O(n \log n)$ operations, and when the points are spread out a search visits $O(\log n)$ boxes plus those that touch the ball. The search of all pairs within D_c (`query_pairs` of `cKDTree` in SciPy [27]) runs this for every point and returns each pair once.

One tree for all the minutes. Phase 1 needs the pairs of points of the *same* minute only. Building one tree per minute would work; a single tree does the same job with a fourth coordinate. Every point gets the coordinates $(X, Y, Z, K\, t_c/60)$, its Earth-centred position and its minute multiplied by $K = 10^8$ m. Two points of the same minute have the same fourth coordinate, so their distance in four dimensions is their distance in space; two points of different minutes are at least $K \gg D_c$ apart and are never returned (Figure 35).

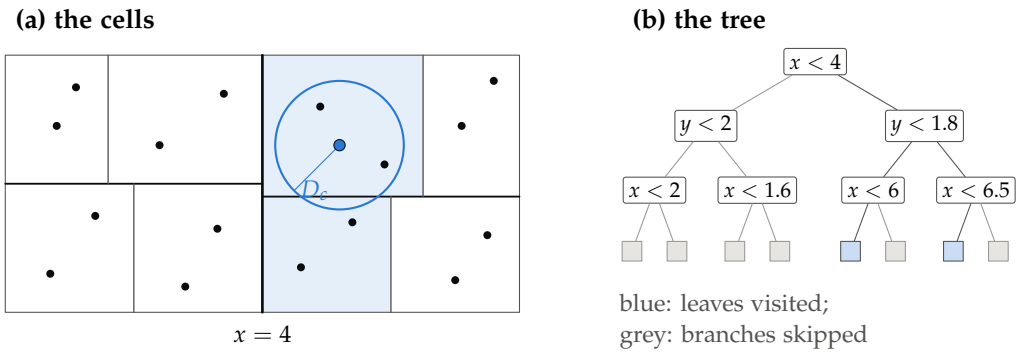


Figure 34: A KD-tree in two dimensions. (a) The sixteen points are split by a vertical line at the median of x, each half by a horizontal line at the median of its y, each quarter by a vertical line again: eight cells of two points. A search of radius D_c around the blue point visits only the two cells that the circle reaches (shaded) and compares their four points; the other twelve are never looked at. (b) The same splits as a tree: the search drops the left branch at the root, since the whole circle lies at $x > 4$, and further down every cell that the circle does not reach.

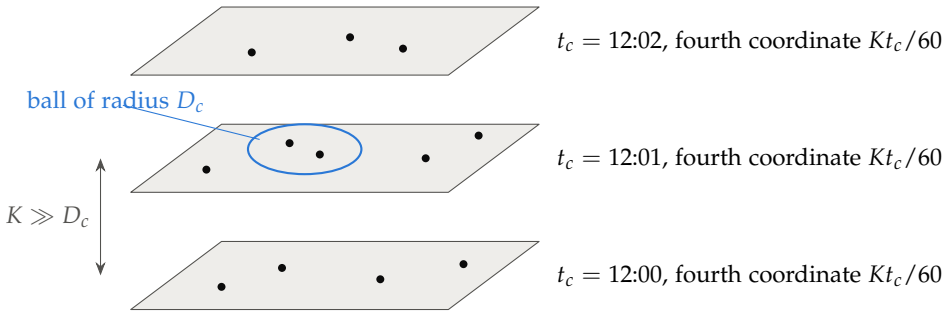


Figure 35: The fourth coordinate. Each minute is a layer of the space $(X, Y, Z, Kt_c/60)$, drawn here as a plane, and the layers are $K \gg D_c$ apart: a search of radius D_c only finds points of the same minute, so one tree does the work of one search per minute.

The steps, for each date.

1. For each piece, one point per whole minute of its extended track, for t_c from $\delta + 60$ s before its first instant to $\delta + 60$ s after its last. This is the interval in which the lemma may ask for the position, whichever of the two roles the flight has.

2. One KD-tree on the four coordinates of all the points of the date.

3. `query_pairs` with radius D_c: pairs of points of the same minute, hence triples (i, j, t_c); the pairs of two pieces of the same flight are dropped.

4. For each candidate pair, with $i < j$, the window is $W_{ij} = \bigcup_{t_c} [t_c - 30, t_c + 30] \cap T_i^{\text{dep}}$, the union over the minutes t_c at which the pair passed the filter: a set of grid instants of the flight of i.

By the lemma, every correspondence $(t, t') \in \mathcal{M}_{ij}$ has $t \in W_{ij}$, and every pair with a correspondence is a candidate. The cost is $O(n_d \log n_d)$ per date, with n_d the number of points of the day.

C.4 Phase 2: exact correspondences

For one candidate pair, Phase 2 tests Definition 3.1 exactly, at every instant t of the window W_{ij} and every delay l of the grid up to δ, and keeps three results: $m_{ij}(t)$, whether i at t has a

correspondence; $m_{ji}(t')$, whether j at t' has one; and $l^*(t)$, the delay of the correspondence of i at t that is closest in 3D (Appendix B).

Lemma C.2. *Scanning the instants $t \in W_{ij}$ and the delays $l \in \Lambda = \{-L\Delta, \dots, L\Delta\}$, with $L = \lfloor \delta/\Delta \rfloor$, enumerates all pairs of $\mathcal{M}_{ij}$; hence m_{ij} and m_{ji} are obtained exactly.*

Proof. Every $(t, t') \in \mathcal{M}_{ij}$ has $t \in W_{ij}$ (Lemma C.1) and $t' - t \in \Lambda$. Moreover $m_{ji}(t') = 1$ if and only if there is a t with $(t, t') \in \mathcal{M}_{ij}$, and every such t lies in W_{ij}. □

Algorithm 1 Phase 2: scan of one candidate pair

Input: flights $i < j$, on the grid; window $W \subseteq T_i^{\text{dep}}$; delays Λ
Output: contact a on the support $W \cup (W + \Lambda)$; delay l^* on W

$m_{ij} \leftarrow 0$; $m_{ji} \leftarrow 0$; $r^{\min} \leftarrow +\infty$; $l^* \leftarrow$ undefined
for $l \in \Lambda$ **do**
  $\tau \leftarrow \{t \in W : \ t + l \in T_j^{\text{dep}}\}$ ▷ array of instants
  $d \leftarrow \|\mathbf{x}_i(\tau) - \mathbf{x}_j(\tau + l)\|$; $h \leftarrow |z_i(\tau) - z_j(\tau + l)|$; $r \leftarrow \sqrt{d^2 + h^2}$
  $c \leftarrow (d < \varepsilon_h) \wedge (h < \varepsilon_z) \wedge D_{ij}(\tau, \tau + l)$
  $m_{ij}[\tau] \leftarrow m_{ij}[\tau] \vee c$; $m_{ji}[\tau + l] \leftarrow m_{ji}[\tau + l] \vee c$
  $u \leftarrow c \wedge (r < r^{\min}[\tau])$; $r^{\min}[\tau] \leftarrow r$ and $l^*[\tau] \leftarrow l$ where u
$a(t) \leftarrow m_{ij}(t) \vee m_{ji}(t)$
return a, l^*

In words. Each pass of the loop fixes one delay l and compares i at every instant of the window with j at the instant l later: one distance, one height difference and, if both glide, one angle per instant. Where the three conditions hold, i at t and j at $t + l$ both get a correspondence. A running minimum of the 3D distance over the delays that pass gives $l^*(t)$.

In the (t, t') plane a pass is the diagonal $t' = t + l$, evaluated over the whole window at once (Figure 36). The code evaluates all the delays of up to 2 000 instants at once, as a matrix of instants by delays; the result is the same as the loop. The cost is $\sum_{\text{pairs}} |W_{ij}| \, (2L + 1)$ distance evaluations, with $2L + 1 = 121$ for $\delta = 5$ min and $\Delta = 5$ s, 49 for $\delta = 2$ min. The memory is $O(|W_{ij}|)$ per pair: boolean arrays are accumulated, without generating the list of pairs of points.

C.5 Phase 3: pair intervals

Phase 3 turns the contact of a pair into the intervals of Definition 4.1.

1. The contact is restricted to the time from the later of the two departures to the earlier of the two landings, so that every edge of the graph (6) joins two nodes of $V(t)$. This cuts at most δ at each end of a contact, where m_{ji} could hold at an instant at which i has not yet departed or has already landed.

2. The instants with $a_{ij} = 1$, sorted, are cut wherever two consecutive ones are separated by a gap of at least g; each run becomes a half-open interval $[t_{\text{on}}, t_{\text{off}})$, with t_{off} one grid step after its last instant.

3. The intervals are kept with their duration. The threshold τ^* is applied after the calibration (Section C.1), and only intervals lasting at least τ^* enter Phase 4.

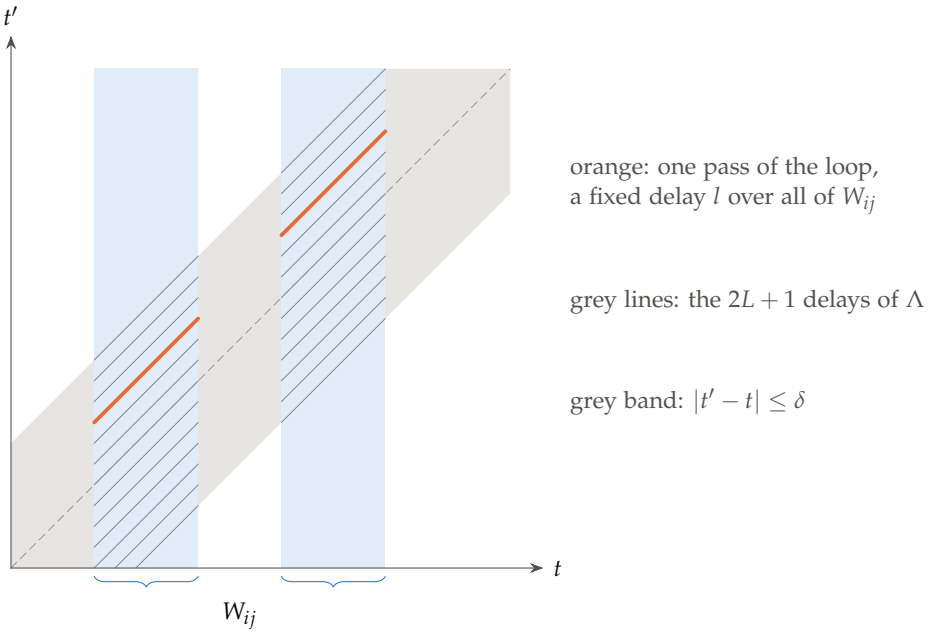


Figure 36: Phase 2 in the (t, t') plane. Only the columns of the window W_{ij} are scanned, one diagonal $t' = t + l$ at a time; by Lemma C.2 this covers all of $\mathcal{M}_{ij}$.

For each interval one also records:

- the signed median delay ℓ_{med}, which tells who follows whom and does not enter Phase 4 (Appendix B);
- the distance between the takeoffs $d_{\text{dec}} = \|\mathbf{x}_i(k_i^0) - \mathbf{x}_j(k_j^0)\|$, which allows requirement (R1) to be imposed afterwards, if wanted.

C.6 Phase 4: groups, configurations and episodes

Phase 4 needs the components of the active graph $G(t)$ at every instant, but $G(t)$ changes only when an interval starts or ends. Between one event and the next the set of active edges is fixed, so the components are recomputed only at the events, and only around the nodes involved (Figure 37).

> **In words.** The events of the day, the starts and ends of the intervals, are sorted in time. At each event the touched flights and the open groups they belong to form a set R; the components inside R are recomputed. An open group whose members no longer form a component is closed; a new component opens a group, linked by a union–find to every group just closed that shares a member with it. A component whose edges change closes its configuration and opens a new one, in the same group if its members did not change.

Lemma C.3. *After the update, the connected component of every node of R is contained in R; hence the local recomputation of Algorithm 2 gives the same components as a global one.*

Proof. Before the update the open groups are disjoint, and those that touch U lie entirely in R. A path that starts in R uses old edges, which stay inside the old component, or new edges, whose endpoints lie in $U \subseteq R$. By induction on the length of the path, every node it reaches is in R. □

Algorithm 2 Phase 4: sweep over the events of one day

Input: pair intervals $\mathcal{I}$ of the day
Output: groups; configurations; episodes (classes of a union–find)

events $\leftarrow \{(t_{\text{on}}, +, \{a, b\}),\ (t_{\text{off}}, -, \{a, b\})\}$ sorted in time
$A \leftarrow \emptyset$ (active edges); $\mathcal{O} \leftarrow \emptyset$ (open groups)
for each instant t_e, with its batch of events B **do**
    $U \leftarrow$ endpoints of the edges in B ▷ nodes involved
    $R \leftarrow U \cup \bigcup \{M_g :\ g \in \mathcal{O},\ M_g \cap U \neq \emptyset\}$
    update A with the events of B
    $\mathcal{K} \leftarrow$ connected components of $(R, A|_R)$ with at least two nodes ▷ local visit
    $X \leftarrow \{g \in \mathcal{O} :\ M_g \subseteq R,\ M_g \notin \mathcal{K}\}$; close every $g \in X$ with $t_g^{\text{end}} \leftarrow t_e$
    for $K \in \mathcal{K}$ that is not already an open group **do**
        open g' with $M_{g'} = K$ and $t_{g'}^{\text{start}} = t_e$
        for $g \in X$ with $M_g \cap K \neq \emptyset$ **do**
            Union(g, g')
    close the open configurations in R whose edges changed; open the new ones in their groups

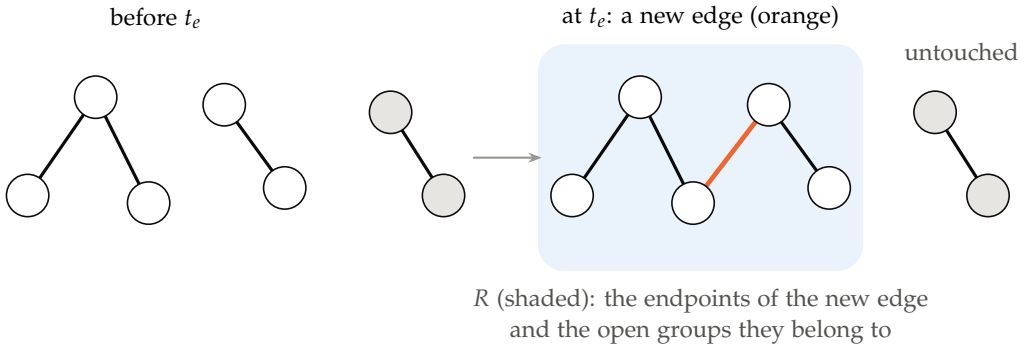


Figure 37: One step of the sweep. Only the nodes of R are visited again; the grey component far away keeps its configuration and its group. The animation of Figure 15 runs the sweep over a whole day.

The cost per day is $O(|\mathcal{I}| \log |\mathcal{I}|)$ for the sorting, plus the sum of the sizes of the components involved at each event, at most a few hundred nodes. The classes of the union–find [26] are the episodes. The dependency blocks follow from a second union–find, which joins the flights that share an episode; a flight never in a group is a block on its own. From the list of the groups one obtains $\gamma_i(t)$ as in (7), filling with S the stretches after the departure not covered by a group and with L the local phase.

C.7 Costs and space

Phase 2 dominates the cost. Phase 1 is worth running on the whole archive first: the number of candidate pairs and the total duration of their windows estimate the cost of Phase 2 before it runs.

D Outputs

What does a run leave, and how is it read? Six tables in Parquet (Table 12); the main one, T3, cuts each flight into stretches of constant membership.

Phase	Cost	Data produced
0	linear in the number of fixes ($\sim 10^9$)	grid points, $\sim 2 \times 10^8$ rows
1	$O(n_d \log n_d)$ per date, $\sum_d n_d \sim 1.7 \times 10^7$	candidate pairs and windows
2	$\sum \lvert W_{ij} \rvert (2L+1)$; measured after Phase 1	temporary arrays per pair
3	linear in the duration of the windows	T1, pair intervals
4	$O(\lvert \mathcal{I} \rvert \log \lvert \mathcal{I} \rvert)$ per date	T2–T5

Table 11: Costs and space. The numbers assume 1.86×10^5 flights of 1.5 h on average and are to be checked on the data.

Table	Columns (type) and content
T1 `pair_intervals`	`date`, `a`, `b` (int32), `t_on`, `t_off` (int32, s), `duration` (int32), `ell_med` (int16, s, signed: positive if `b` follows `a`), `d_dec` (float32, m). Who was with whom, and when.
T2 `groups`	`date`, `group_id`, `episode_id` (int32), `t_start`, `t_end` (int32), `n` (int16), `members` (list of int32).
T3 `membership`	`flight` (int32), `t_start`, `t_end` (int32), `state` (int8: L, S, G), `group_id`, `episode_id` (int32, null if L or S), `n` (int16, 1 if S). The main table: each flight cut into stretches of constant membership.
T4 `episodes`	`episode_id`, `date`, `t_start`, `t_end`, `n_flights`, `n_max`, `n_groups`, `block`.
T5 `flight_summary`	`flight`, `phi_G` (fraction of T_i^{dep} in a group), `kappa` (distinct companions), `n_entries`, `n_episodes`, `block`, `local` (boolean).
T6 `configurations`	`date`, `config_id`, `group_id` (int32), `t_start`, `t_end` (int32), `n` (int16), `members` (list of int32), `edges` (list of pairs of int32). How the members of a group are linked, between two changes of the edges.

Table 12: Final tables, in Parquet. The members of a group are not repeated in T3: they are read from T2 through `group_id`.

The state of flight i at instant t is obtained with an as-of join on (`flight`, t) in T3, sorted by `t_start` (Figure 38). T3 has about $N_{\text{flights}}(2 + 2\bar{g})$ rows, with $\bar{g}$ the mean number of groups per flight: of the order of 10^5–10^6 rows, tens of MB. The other tables are smaller.

E Why $m_{ij}(t) = 1$ does not give $m_{ji}(t) = 1$

$m_{ij}(t)$ of Definition 3.2 asks whether i, at instant t, is at a place that j visits within δ; $m_{ji}(t)$ asks whether j, at the same instant, is at a place that i visits within δ. The first fixes the instant of i and lets that of j vary within $\pm\delta$; the second does the opposite. Both are read from one set of correspondences $\mathcal{M}_{ij}$, one on each axis, and they agree only if that set is symmetric about the diagonal $t' = t$. What does hold:

- if $m_{ij}(t) = 1$ through the pair (t, t'), then $m_{ji}(t') = 1$: the same pair, read on the other axis, $l = t' - t$ later;
- every instant of one indicator lies within δ of an instant of the other;
- the contact $a_{ij} = m_{ij} \vee m_{ji}$ is symmetric: $a_{ij} = a_{ji}$.

This appendix keeps only the horizontal condition of $\mathcal{M}_{ij}$: the vertical and heading conditions only remove pairs.

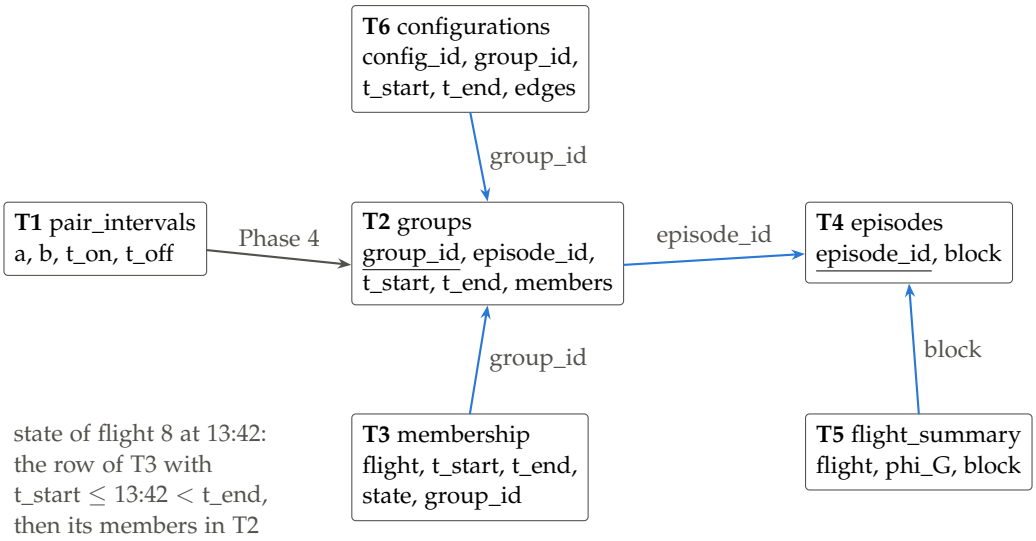


Figure 38: How the tables connect. Arrows point from a column to the table where it is a key. The example reads the state of a flight at one instant with an as-of join on T3.

E.1 A worked example

Take $\delta = 120$ s and $\varepsilon_h = 500$ m, and ignore heights and headings. Pilot i circles in a thermal at the point P from 0 to 600 s; the radius of the circle is negligible against ε_h. Pilot j flies east in a straight line at 10 m/s and is in P at 500 s.

t (s)	j relative to P	m_{ij}	m_{ji}	a_{ij}	what happens
300	2.0 km west	0	0	0	j needs more than 120 s to come within 500 m of P
380	1.2 km west	1	0	1	i is where j will be 120 s later; j is not yet where i has been
500	in P	1	1	1	both are in P
560	0.6 km east	1	0	1	i is where j was 60 s earlier; j has left

Table 13: The four instants of the example. Over the whole flight $m_{ij} = 1$ from 331 to 600 s, $m_{ji} = 1$ from 451 to 549 s, and $a_{ij} = 1$ from 331 to 600 s (strict inequality: exactly 500 m does not count).

E.2 The same example in the plane (t, t')

Plot every correspondence (t, t') with the instant at which i passes across and the one at which j passes up (Figure 40a). Then $m_{ij}(t) = 1$ exactly when the *vertical* line at t meets $\mathcal{M}_{ij}$, and $m_{ji}(t) = 1$ exactly when the *horizontal* line at height t meets it. The two indicators are the shadows of $\mathcal{M}_{ij}$ on the two axes.

At $t = 380$ the vertical line meets $\mathcal{M}_{ij}$ (at the pair $(380, 500)$ and the ones just below it), and the horizontal line does not: $\mathcal{M}_{ij}$ has no point with $t' < 451$. The set is not symmetric about the diagonal. The mirror of the pair $(380, 500)$ would be $(500, 380)$, which needs i at 500 s within 500 m of j at 380 s, and that is 1.2 km away.

E.3 How far apart they can be

Claim. If $m_{ij}(t) = 1$ there is an instant t' with $|t' - t| \le \delta$ and $m_{ji}(t') = 1$; and conversely.

Proof. Take the pair $(t, t') \in \mathcal{M}_{ij}$ that gives $m_{ij}(t) = 1$. Then $|t' - t| \le \delta$ by the definition of $\mathcal{M}_{ij}$, and $m_{ji}(t') = 1$ by the definition of m_{ji}. The converse is the same argument read the other way. □

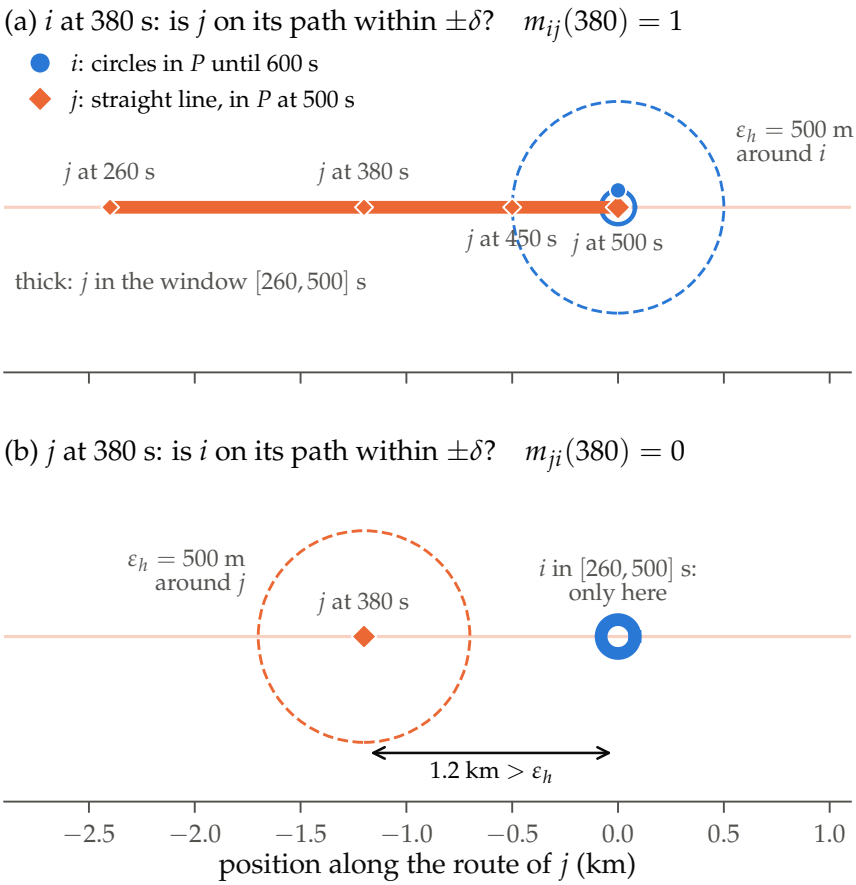


Figure 39: The example at 380 s. (a) i is in P and j is 1.2 km west of it. In the window $[260, 500]$ s of $\pm\delta$ (thick), j comes within ε_h of i at 451 s and is in P at 500 s: the pair $(380, 500)$ is a correspondence, so $m_{ij}(380) = 1$. (b) At the same instant j is 1.2 km from P, and during its window i has been only in P: no position of i lies within ε_h of j, so $m_{ji}(380) = 0$. The ring only marks P.

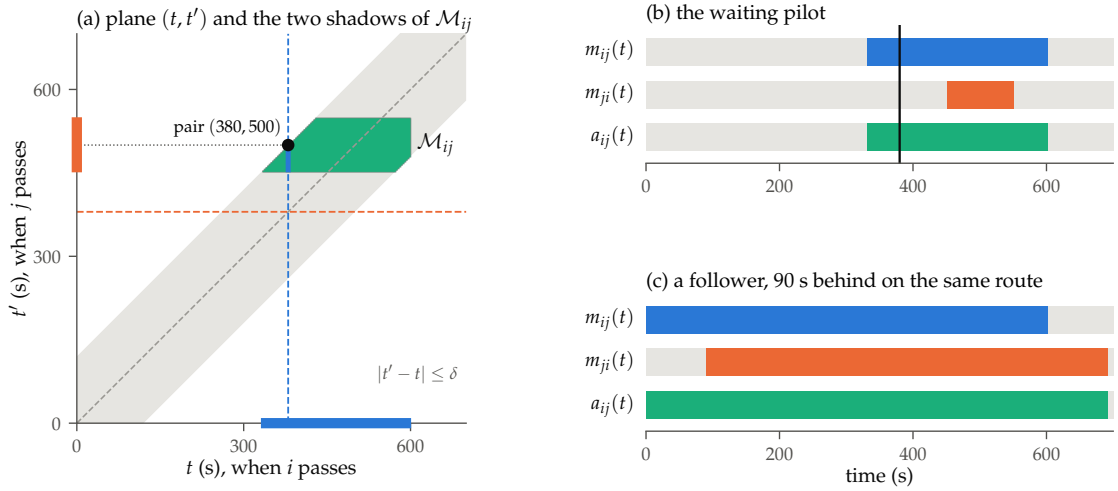


Figure 40: (a) The correspondences $\mathcal{M}_{ij}$ (green) in the band $|t' - t| \le \delta$, and its shadows m_{ij} (blue bar on the t axis) and m_{ji} (orange bar on the t' axis). The vertical line at 380 s meets $\mathcal{M}_{ij}$ (thick segment), the horizontal one does not. (b) The indicators over time, with the line at 380 s. (c) A pilot j on the same route 90 s behind i: m_{ji} is m_{ij} delayed by exactly 90 s.

So the two indicators can differ only by a shift of at most δ. The bound is reached in the example: the first instant of m_{ij} (331 s) is exactly δ before the first instant of m_{ji} (451 s). The union a_{ij} inherits it: at each end of a contact it extends the shorter indicator by up to δ.

E.4 Why the contact is the union

definition	symmetric	in the example	remark
$m_{ij} \vee m_{ji}$ (used)	yes	331–600 s	counts approach and departure within δ
$m_{ij} \wedge m_{ji}$	yes	451–549 s	shorter: each on the path of the other at once
m_{ij} alone	no	331–600 s, or 451–549 s for (j, i)	depends on which pilot is called i

Table 14: Three ways to define the contact of a pair.

The union makes the contact independent of the order of the two pilots and keeps the approach and the departure, where one of them is already on the path of the other. The price is that a contact starts up to δ before the pilots are close at the same time. The mutual version $m_{ij} \wedge m_{ji}$ would give shorter contacts: the false discovery rate and τ^* would have to be computed again with the null model, which is not done here.

In the code. `contacts.scan` returns the instants of $m_{ij} = 1$ and, separately, those of $m_{ji} = 1$; `contacts.pair_intervals` takes their union. Scenarios: `models.WAIT`, `models.FOLLOW`; figures: `plots.asymmetry`; numbers checked in `test_note.py`.

F Graphs, from scratch

The few terms of graph theory that the note uses, each with the place where it is used.

- **Graph.** A set of *nodes* and a set of *edges*, each edge joining two nodes. The active graph $G(t)$ of Section 4.3 has a node for each flight and an edge for each pair interval in progress.
- **Path.** A sequence of nodes, each joined by an edge to the next. Two nodes are *connected* when a path joins them.
- **Connected component.** A largest set of nodes connected to each other. Every node is in exactly one component; a node with no edge is a component alone (a pilot flying solo). A configuration is a component with at least two nodes, with its edges.

Two members of a configuration need not share an edge: in the chain 1–2–3 of Figure 41, pilots 1 and 3 are in the same configuration through pilot 2.

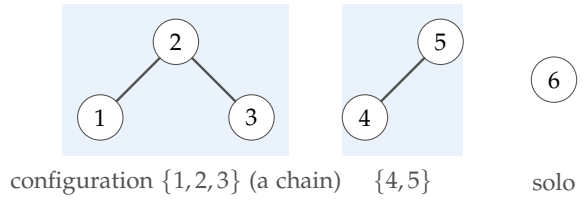


Figure 41: The graph at one instant: three components, two of them configurations.

- **Directed graph.** The edges carry an arrow. For the episodes the nodes are groups, with an arrow $g \to g'$ when g' starts as g ends and they share a member (Section 4.4). An episode is a connected component of this graph, arrows ignored.

- **Bipartite graph.** Nodes of two kinds, edges only between kinds. At one instant, the groups ending and those starting, with an edge when they share a member. Each component is one event (Figure 42); its outcome follows from how many groups it has on each side: one to one is a join or a leave (members added or removed), one to several a split, several to one a merge (Section 7.1).

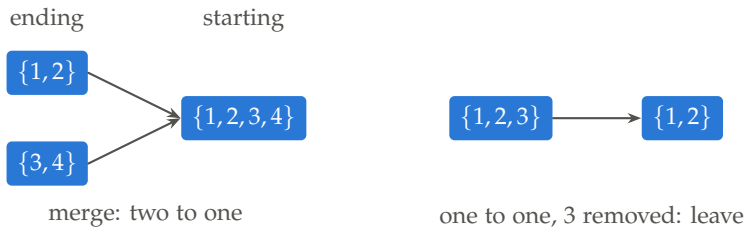


Figure 42: Two events: each is a connected component of the bipartite graph of the groups ending and starting at one instant.

Shape (Section 7.3). For a connected graph of n nodes and m edges:

- **Density** $m/\binom{n}{2}$: the share of the node pairs joined by an edge.
- **Distance** between two nodes: the fewest edges of a path joining them. **Diameter**: the largest distance. In the chain 1–2–3 the diameter is 2; in a triangle, 1.
- **Complete graph**: every pair joined, density 1, diameter 1. **Tree**: $m = n - 1$, the fewest edges that keep n nodes connected, with no closed path; a *chain* is a tree with diameter $n - 1$, a *star* one with diameter 2.

Order (Section 7.3). In a directed graph:

- a **cycle** is a path that follows the arrows back to its start (A→B→C→A); a graph with none is **acyclic**, and its nodes can be put in a line with every arrow pointing forward;
- a **source** has no arrow coming in: here, a pilot with nobody ahead, a leader;
- the **depth** of an acyclic graph is its longest chain of arrows: 1 for a star, $n - 1$ for a line.

A directed triangle is *transitive* (A→B, B→C, A→C) or *cyclic*; with arrows drawn at random, 2 of its 8 orientations are cyclic.

Configuration model (Section 7.4). A random graph with the *degrees* (the number of edges of each node) of a given one: each node gets as many half-edges as its degree, and the half-edges are paired at random [21]. It keeps how many partners each pilot had and changes only who they were.

G Two models

Section 7 reads two measured quantities through two textbook models.

Ornstein–Uhlenbeck process [29] (Section 7.2). A quantity ℓ pulled back to 0 by a spring of rate k and shaken by noise:

$$d\ell = -k\,\ell\,dt + \sqrt{2k}\,\sigma\,dW. \tag{19}$$

It forgets its value after $1/k$ and spreads by σ around 0. Its mean square change after a lag s is $D(s) = \langle(\ell(t+s) - \ell(t))^2\rangle = 2\sigma^2(1 - e^{-ks})$: it grows as $2k\sigma^2 s$ at short lags, like a free random walk, and stops at $2\sigma^2$ after $1/k$. With $k = 0$ it is a random walk and D grows forever. A measurement noise of variance $c/2$ on each value adds c to $D(s)$ at every lag.

Birth–death chain (Section 7.5). A group of n pilots gains one at rate $g(n)$ and loses one at rate $l(n)$. Over a long time the transitions from n to $n+1$ and back are equally many, so the times spent at the two sizes satisfy $T(n)\,g(n) = T(n+1)\,l(n+1)$, Eq. (13), and multiplying the ratios from 2 to n gives $T(n)$. If every member brings in others at rate a and leaves at rate μ, then $g(n) = an$, $l(n) = \mu n$ and $T(n) \propto x^n/n$ with $x = a/\mu$, the log-series, whose tail falls exponentially. A leaving rate that falls with n makes the tail heavier.

H Further evidence

These measurements support Sections 5–7 without changing their conclusions.

The time shifts of the surrogates (Section 5.1). Figure 43 gives the whole distribution behind the median of 58 min.

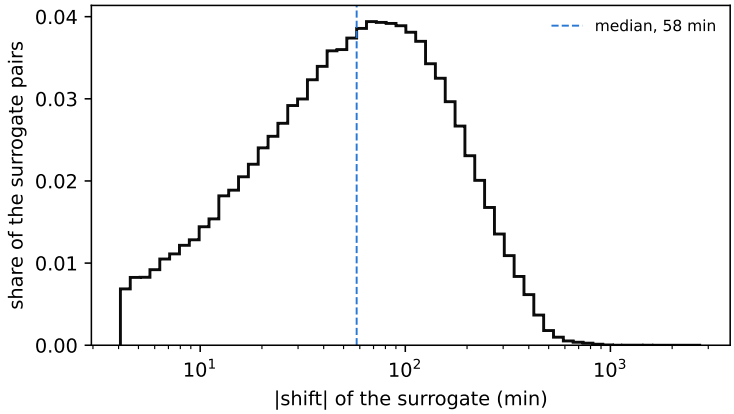


Figure 43: The time shifts $|r_{ij}|$ of the 248 741 surrogate pairs of the chosen run, whole archive.

The delay inside the contacts (Section 6.1). Figure 44 gives the whole distribution behind Table 5: short intervals spread over the window and pile up at its edge, long ones gather at small delays.

The distance at the same instant (Section 6.1). Is there a distance beyond which two pilots stop being closer than chance? We counted the distance between the two pilots at every grid instant when both fly after departure, without requiring a correspondence, for all the real pairs and for the surrogates, on the dates of Figure 18 that have surrogates (210 of 300, 58 901 h of real exposure). Real pairs are much closer than chance at short distance and have no edge (Figure 45): the ratio $g(d)$ is 10.4 within 100 m, below 2 from 500 m, below 1.5 from 900 m, below 1.1 from 2.7 km and below 1 from 5.7 km. Real pairs spend 6.6% of their time together

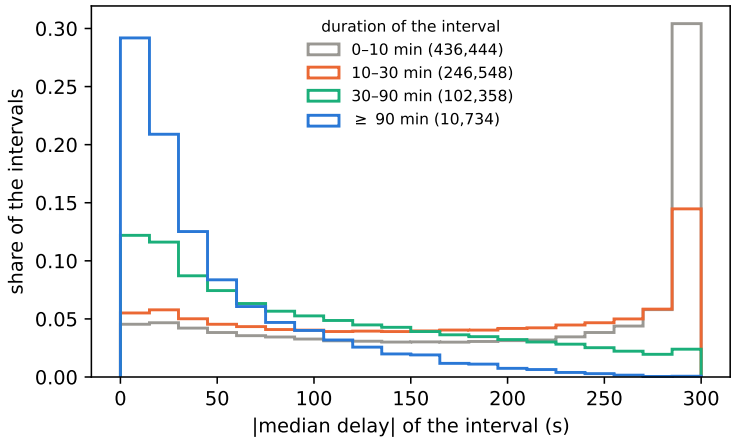


Figure 44: Distribution of the $|\ell_{\text{med}}|$ of the pair intervals, by their duration ($\delta = 5$ min, whole archive).

within 1 km of each other, against 2.8% by chance. Beyond 6 km g falls slightly below 1, because both distributions sum to one.

The excess reaches several kilometres, beyond the visual range of 1.5 km of Section 3.4. Since the surrogates also change the hour, this excess may hold the weather of the hour as well as sight, so it bounds neither the visual range nor the range of following.

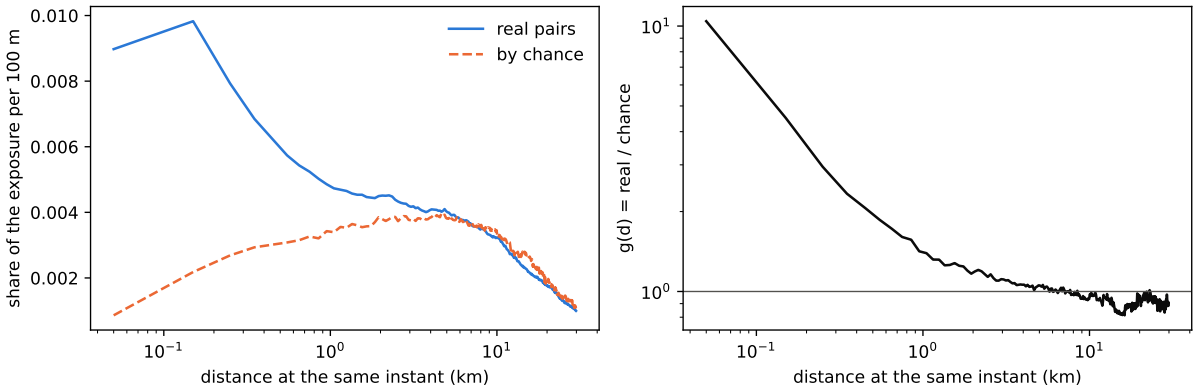


Figure 45: Distance between the two pilots at the same instant, over all the time both fly, on 210 dates. Left: share of the exposure in each 100 m bin, real pairs and chance. Right: their ratio $g(d)$.

One parameter at a time (Section 6.3). Table 15.

Phase at entry and exit, time of day, habitual leaders (Section 7). Table 16, Figure 46 and Figure 47.

References

[1] Z. Ákos, M. Nagy, T. Vicsek. Comparing bird and human soaring strategies. *Proceedings of the National Academy of Sciences* 105(11):4139–4143, 2008.

[2] C. Amornbunchornvej, I. Brugere, A. Strandburg-Peshkin, D. R. Farine, M. C. Crofoot, T. Y. Berger-Wolf. Coordination event detection and initiator identification in time series data. *ACM Transactions on Knowledge Discovery from Data* 12(5), 2018. doi:10.1145/3201406.

Setting ($\delta = 5$ min unless stated)	τ^* (min)	f_G	episodes	flights in a group
starting values	126.1	0.034	98	210
$\varepsilon_h = 250$ m	75.1	0.058	253	537
$\varepsilon_h = 1000$ m	126.8	0.065	162	372
$\varepsilon_z = 150$ m	126.1	0.020	61	125
$\varepsilon_z = 600$ m	126.1	0.044	122	268
$\delta = 2$ min	42.6	0.106	600	1 170
$\delta = 10$ min	128.4	0.062	153	365
$\theta_{\max} = 30°$	126.1	0.034	97	208
$\theta_{\max} = 90°$	126.1	0.034	98	210
$g = 60$ s	75.8	0.087	332	732
$g = 300$ s	126.1	0.055	141	318

Table 15: One parameter at a time on 300 random dates (7 051 flights), $q = 0.05$.

	climb (%)	search (%)	transition (%)
entering a group	28.1 [27.6, 28.5]	10.3 [10.0, 10.6]	61.5 [61.0, 61.9]
leaving a group	28.0 [27.6, 28.3]	11.2 [10.9, 11.5]	60.6 [60.1, 61.0]
share of time, alone	29.1 [28.9, 29.3]	10.1 [10.0, 10.2]	60.8 [60.6, 61.0]
share of time, in a group	33.0 [32.8, 33.2]	9.4 [9.2, 9.6]	57.6 [57.3, 57.9]

Table 16: Phase at the instant of entering or leaving a group, and shares of time in each phase (percent, with 5–95% bands; 1 Hz flights).

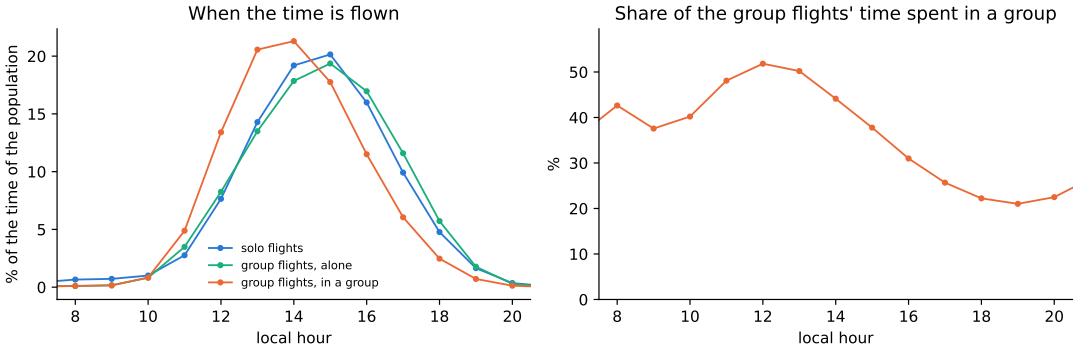


Figure 46: Time of solo flights (S), of group flights alone (A) and in a group (G) over the local hour (left), and the share of a group flight's time spent in a group (right).

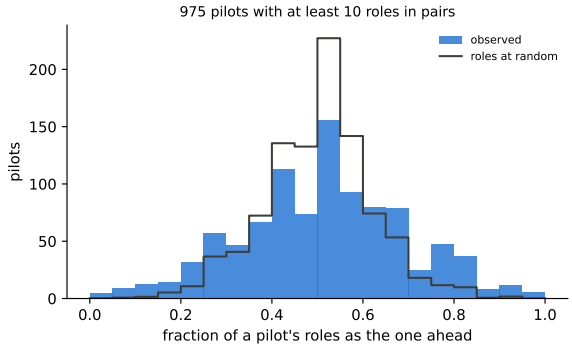


Figure 47: Lead fraction of each pilot, against roles at random.

[3] C. Amornbunchornvej, T. Y. Berger-Wolf. Framework for inferring leadership dynamics of complex movement from time series. *Proceedings of the SIAM International Conference on Data Mining*, 549–557, 2018.

[4] C. Amornbunchornvej. mFLICA: an R package for inferring leadership of coordination from time series. *SoftwareX*, 2021. doi:10.1016/j.softx.2021.100781. Preprint arXiv:2004.06092.

[5] M. Andersson, J. Gudmundsson, P. Laube, T. Wolle. Reporting leadership patterns among trajectories. *Proceedings of the ACM Symposium on Applied Computing*, 3–7, 2007.

[6] E. Bonabeau, L. Dagorn, P. Fréon. Scaling in animal group-size distributions. *Proceedings of the National Academy of Sciences* 96:4472–4477, 1999.

[7] K. Buchin, M. Buchin, M. van Kreveld, B. Speckmann, F. Staals. Trajectory grouping structure. *Journal of Computational Geometry* 6(1), 2015. doi:10.20382/jocg.v6i1a3. Preliminary version: WADS 2013; arXiv:1303.6127.

[8] L. Chen, M. T. Özsu, V. Oria. Robust and fast similarity search for moving object trajectories. *Proceedings of ACM SIGMOD*, 491–502, 2005.

[9] A. Flack, M. Nagy, W. Fiedler, I. D. Couzin, M. Wikelski. From local collective behavior to global migratory patterns in white storks. *Science* 360(6391):911–914, 2018.

[10] D. Greene, D. Doyle, P. Cunningham. Tracking the evolution of communities in dynamic social networks. *Proceedings of the International Conference on Advances in Social Networks Analysis and Mining (ASONAM)*, 176–183, 2010.

[11] J. Gudmundsson, M. van Kreveld. Computing longest duration flocks in trajectory data. *Proceedings of ACM GIS*, 2006.

[12] H. Jeung, M. L. Yiu, X. Zhou, C. S. Jensen, H. T. Shen. Discovery of convoys in trajectory databases. *Proceedings of the VLDB Endowment* 1(1), 2008.

[13] R. Joo, M.-P. Etienne, N. Bez, S. Mahévas. Metrics for describing dyadic movement: a review. *Movement Ecology* 6:26, 2018.

[14] M. Konzack, T. McKetterick, T. Ophelders, M. Buchin, L. Giuggioli, J. Long, T. Nelson, M. A. Westenberg, K. Buchin. Visual analytics of delays and interaction in movement data. *International Journal of Geographical Information Science* 31(2):320–345, 2017.

[15] Z. Li, B. Ding, J. Han, R. Kays. Swarm: mining relaxed temporal moving object clusters. *Proceedings of the VLDB Endowment* 3(1), 2010.

[16] Z. Li, F. Wu, M. C. Crofoot. Mining following relationships in movement data. *Proceedings of the 13th IEEE International Conference on Data Mining (ICDM)*, 458–467, 2013. doi:10.1109/ICDM.2013.98.

[17] J. A. Long, T. A. Nelson, S. L. Webb, K. L. Gee. A critical examination of indices of dynamic interaction for wildlife telemetry studies. *Journal of Animal Ecology* 83(5):1216–1233, 2014.

[18] J. A. Long, S. L. Webb, S. M. Harju, K. L. Gee. Analyzing contacts and behavior from high frequency tracking data using the wildlifeDI R package. *Geographical Analysis* 54:648–663, 2022.

[19] M. Nagy, Z. Ákos, D. Biro, T. Vicsek. Hierarchical group dynamics in pigeon flocks. *Nature* 464:890–893, 2010.

[20] M. Nagy, I. D. Couzin, W. Fiedler, M. Wikelski, A. Flack. Synchronization, coordination and collective sensing during thermalling flight of freely migrating white storks. *Philosophical Transactions of the Royal Society B* 373:20170011, 2018.

[21] M. E. J. Newman. *Networks: An Introduction*. Oxford University Press, 2010.

[22] W. L. Ruzzo, M. Tompa. A linear time algorithm for finding all maximal scoring subsequences. *Proceedings of ISMB*, 1999.

[23] H. Sakoe, S. Chiba. Dynamic programming algorithm optimization for spoken word recognition. *IEEE Transactions on Acoustics, Speech, and Signal Processing* 26(1):43–49, 1978.

[24] O. Spiegel, S. T. Leu, A. Sih, C. M. Bull. Socially interacting or indifferent neighbours? Randomization of movement paths to tease apart social preference and spatial constraints. *Methods in Ecology and Evolution* 7(8):971–979, 2016.

[25] H. Su, S. Liu, B. Zheng, X. Zhou, K. Zheng. A survey of trajectory distance measures and performance evaluation. *The VLDB Journal* 29:3–32, 2020.

[26] R. E. Tarjan. Efficiency of a good but not linear set union algorithm. *Journal of the ACM* 22(2):215–225, 1975.

[27] P. Virtanen et al. SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nature Methods* 17:261–272, 2020.

[28] V. G. Tusher, R. Tibshirani, G. Chu. Significance analysis of microarrays applied to the ionizing radiation response. *Proceedings of the National Academy of Sciences* 98(9):5116–5121, 2001.

[29] G. E. Uhlenbeck, L. S. Ornstein. On the theory of the Brownian motion. *Physical Review* 36:823–841, 1930.

[30] M. Vlachos, G. Kollios, D. Gunopulos. Discovering similar multidimensional trajectories. *Proceedings of the 18th International Conference on Data Engineering (ICDE)*, 2002.

[31] H. J. Williams, A. J. King, O. Duriez, L. Börger, E. L. C. Shepard. Social eavesdropping allows for a more risky gliding strategy by thermal-soaring birds. *Journal of the Royal Society Interface* 15:20180578, 2018.